

MAGNETOM

Function Description System

Magnet System

Symphony, A Tim System

Document Version

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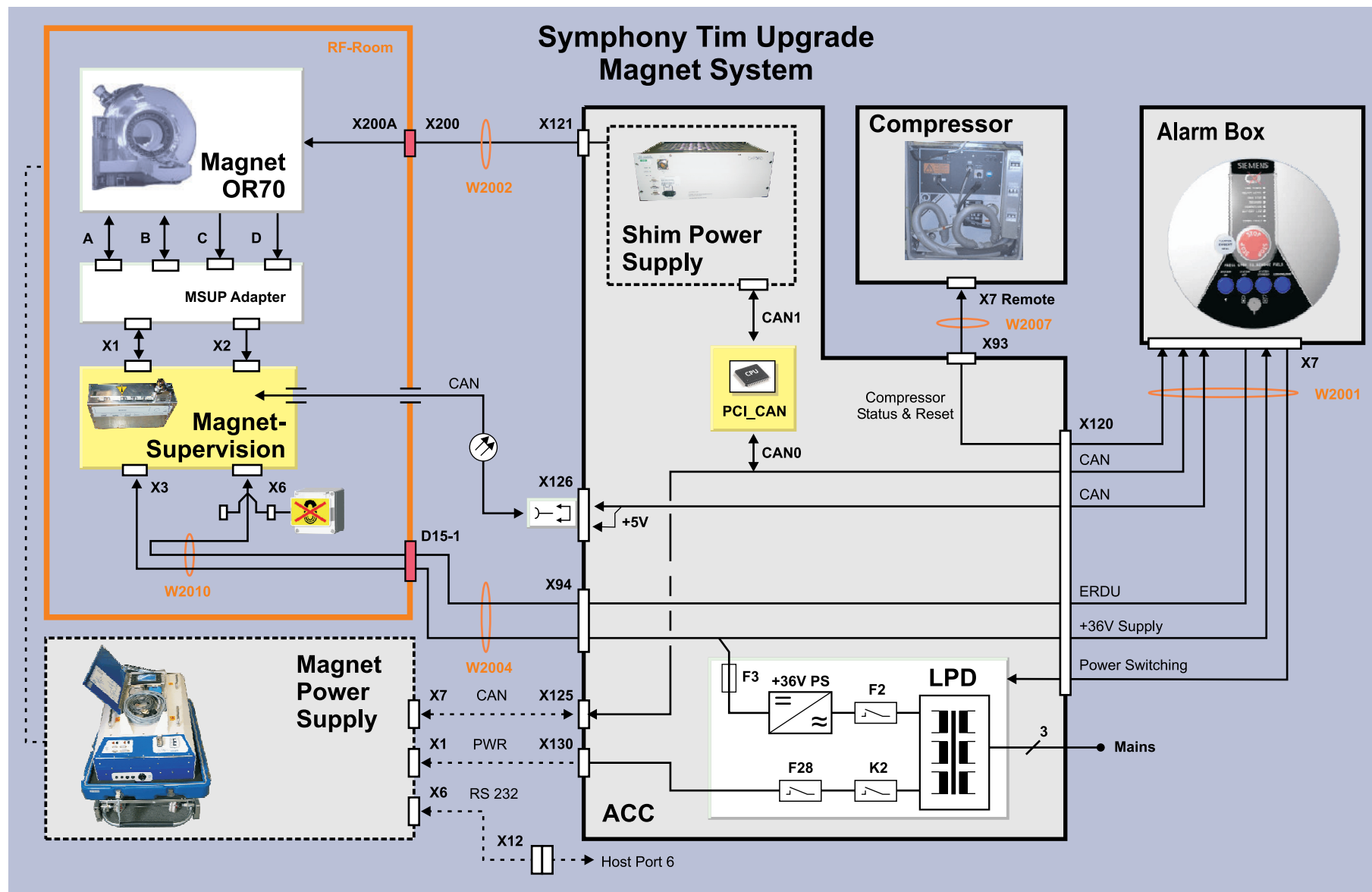
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The magnet system of the 1.5 Tesla MAGNETOM Symphony Tim Upgrade retains the superconducting magnet of the MAGNETOM Symphony - the OR70. Since the "old" CCA cabinet is replaced by a new ACC during the update, an external Magnet Supervision (MSUP), very similar to Avanto/Espree systems, is used with some cable adaptations to the magnet.

To energize the magnet, the MPS3600 Magnet Power Supply that has been introduced with Avanto systems is used for ramping the OR70 magnet, together with a new "MPSU Accessory Kit" and "MPS3600 Standard Extension Kit."

Fig. 1: Magnet System Overview



The following components offering support and monitoring are part of the Symphony Tim Upgrade magnet system:

- **OR70 1.5 Tesla superconducting magnet** - Generates the main magnetic field B_0 . The magnet is passively shimmed during installation to achieve the required field homogeneity.
- **Magnet Monitoring** - Monitoring system for magnet and refrigerator functions.
- **Alarm Box** - Indicator for warning and alarm messages of the Magnet Monitoring. It includes system power switch buttons and a magnet stop button (ERDU).
- **Advanced High Order Shim option** - With the help of the electric shim, the magnetic field homogeneity with the patient can be further improved.
- **Refrigerator System** - Compressor and cold head unit for cooling the cryo shields of the magnet. Two different variants manufactured by Leybold or SHI-APD exist.

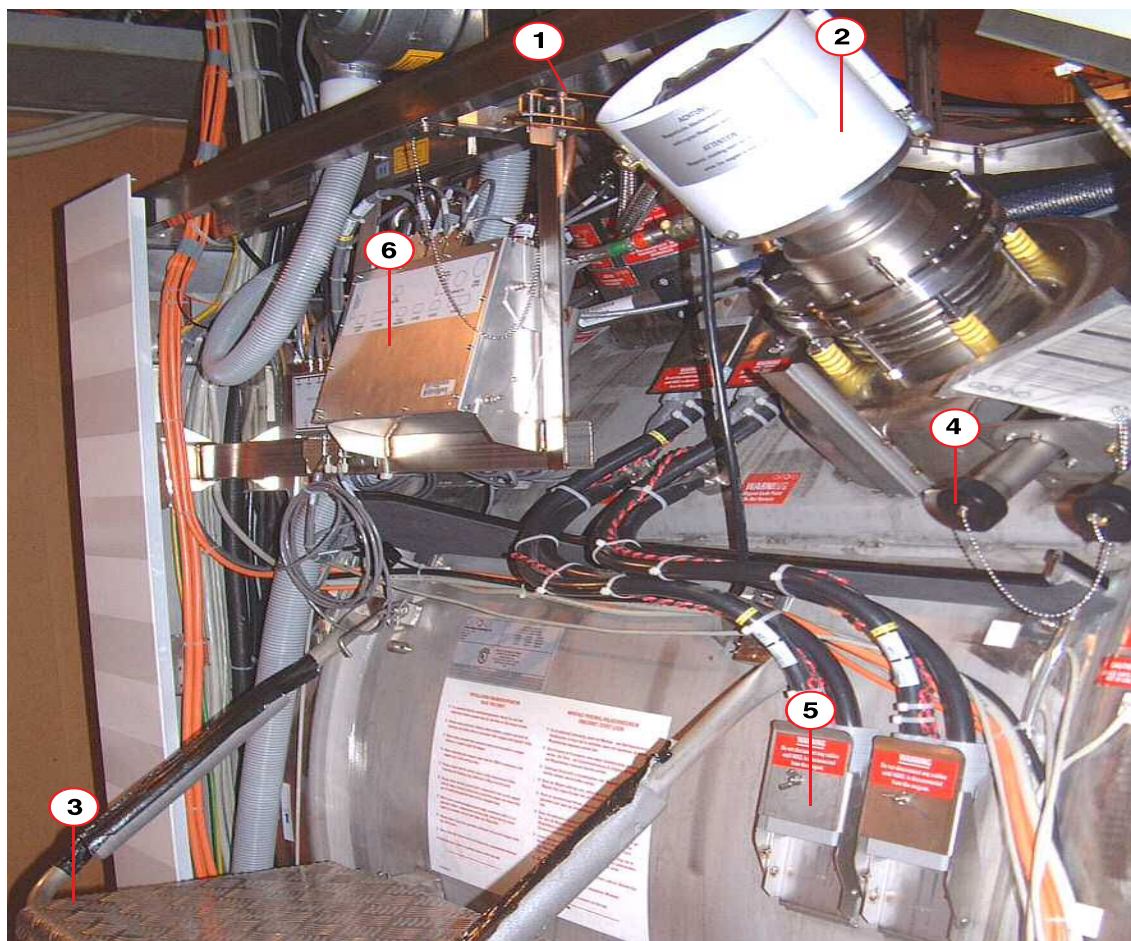
For ramping the OR70 magnet, the following service tools are required:

- **Magnet Power Supply 3600, MPSU Accessory Kit and MPS3600 Standard Extension Kit** - The MPSU Accessory Kit and MPS3600 Standard Extension kit is additionally required to adapt the MPS to the OR70 magnet.

3.1 Overview

3.1.1 General

Fig. 2: Magnet View OR70



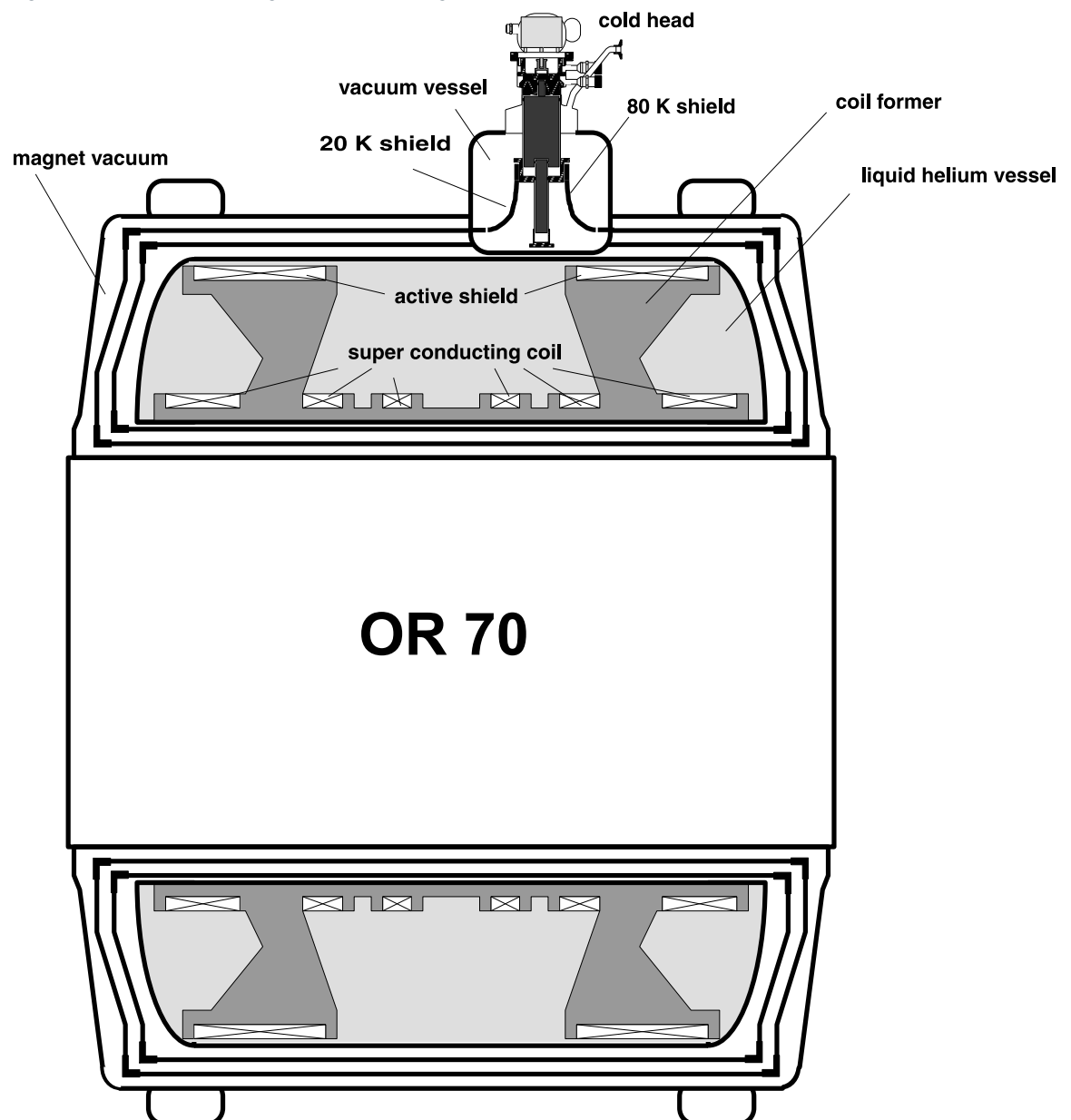
- (1) Service turret (behind cold head)
- (2) Cold head
- (3) Service platform
- (4) Helium fill syphon
- (5) Connectors for Magnet Power Supply
- (6) Magnet Supervision

3.1.2 Function

The Symphony Tim Upgrade magnet OR70 is based on the principle of superconductivity. The actively shielded magnet assembly is sealed into an all welded stainless steel vessel which is filled with liquid helium. This liquid helium vessel is assembled into an all welded, vacuum insulated, stainless steel vacuum chamber, inside of which two aluminium radiation shields reduce heat that is radiated to the liquid helium vessel. The shields are cooled by a two stage Gifford-McMahon refrigerator (see also Refrigerator System)

The cryogenic and the electrical access is provided by an integrated turret.

Fig. 3: Cross-sectional Diagram of OR70 Magnet



3.1.3 Specifications

Tab. 1 Magnet Technical Data

Parameter	OR70
Effective field strength	1.5 T
0.5 mT fringe field distance	Z-dir 4.0 m X,Y-dir 2.5 m
Max. field decay	0.1 ppm/h, 55 kHz/Year
Magnet weight (100% He, without coils and table)	3900 kg
Magnet length	1.6 m
Magnet height (with turret)	2.33 m
Magnet outer diameter	1.97 m
Magnet bore	0.9 m
Imaging volume	500 mm DSV
Ramping time to nominal field	45 min.
Typical current	585 A
Max. voltage during ramping	9.9 V
Stabilization after ramping	≤ 5 ppm /hr after 10 min
100% Helium Level	1710 l
30% helium level (min level)	436 l
Maximum He boil-off	0.075 l/h
Typical He boil off	0.050 l/h

3.2 Magnet Coil Components

3.2.1 General

SMT active shield magnets are very similar in construction. The OR70 magnet has 6 inner coil journals and 2 outer coils that provide the active shield (i.e. reducing the magnetic fringe field).

3.2.2 Function

3.2.2.1 Main Coil

The superconducting magnet coil is designed to generate a homogeneous, stable main magnetic field B_0 at a minimum fringe field.

3.2.2.2 Superconductive Switch

A superconductive switch, actuated by a switch heater, allows the magnet to be energized and de-energized.

3.2.2.3 Quench Heaters

In case of emergency (accidents, fire, etc.), the magnetic field has to be de-energized immediately to avoid danger to personnel. Two quench heaters are placed close to the main coil. When activated, they immediately turn the magnet wire into the normal-conductive state (Quench). The coil current is discharged by the wire resistance within 20 seconds. The thermal energy released will boil off a significant amount of liquid helium.

3.2.2.4 Quench Protection Diodes

During a quench, up to 6 kV can be induced across the coil. To avoid damage, the quench protection diodes clamp the voltage to a maximum of approximately 12 V.

3.2.2.5 External Interference Screen (EIS)

On top of the coil journal, superconductive External Interference Screen (EIS) coil windings are placed. This coil picks up transient B_0 interference and shields the main coil from magnetic field fluctuation caused by moving cars, elevators, power lines, etc..

Over time, the main magnetic field decays - caused by the rest resistance of the coil joints. Hence, a rising current is induced into the EIS, producing a counter-field that deteriorates B_0 field homogeneity. Therefore, the EIS has to be quenched once a day by a timer logic of the Magnet Supervision Unit (MSUP).

To avoid image quality problems, the EIS has to be reset when the system is not utilized (typical time set-up = 2:00 am).

3.3 Service Turret

3.3.1 General

Fig. 4: Service turret



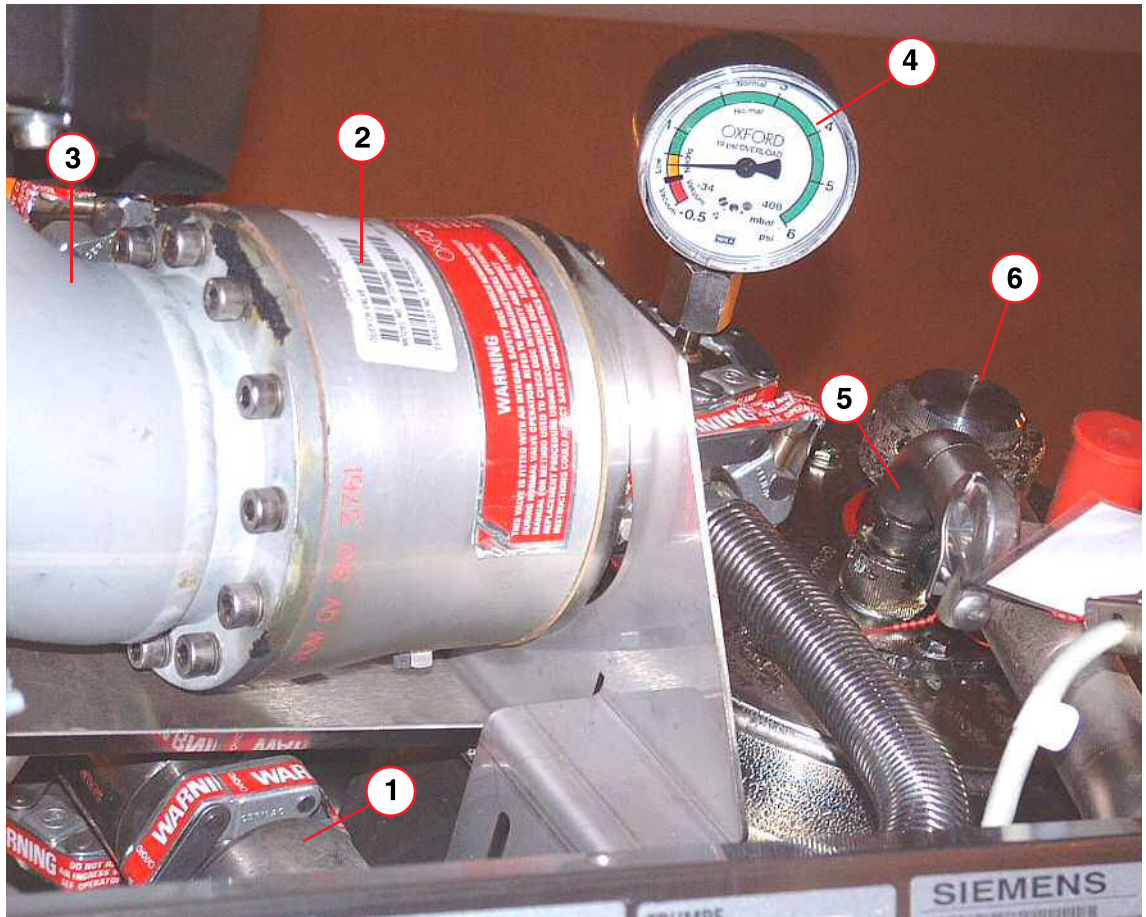
- (1) Venting system
- (2) Entry port for helium fill syphon
- (3) Entry port for Enhanced Current Lead (ECL)
- (4) Entry port for Enhanced Current Lead (ECL)

(4) Electrical interfaces to the Magnet Supervision (MSUP)

3.3.2 Function

3.3.2.1 Venting System

Fig. 5: Venting system OR70



- (1) Vent valve
- (2) Quench valve
- (3) Quench tube
- (4) Pressure gauge
- (5) Helium fill syphon
- (6) Service port for Enhanced Current Lead (ECL)

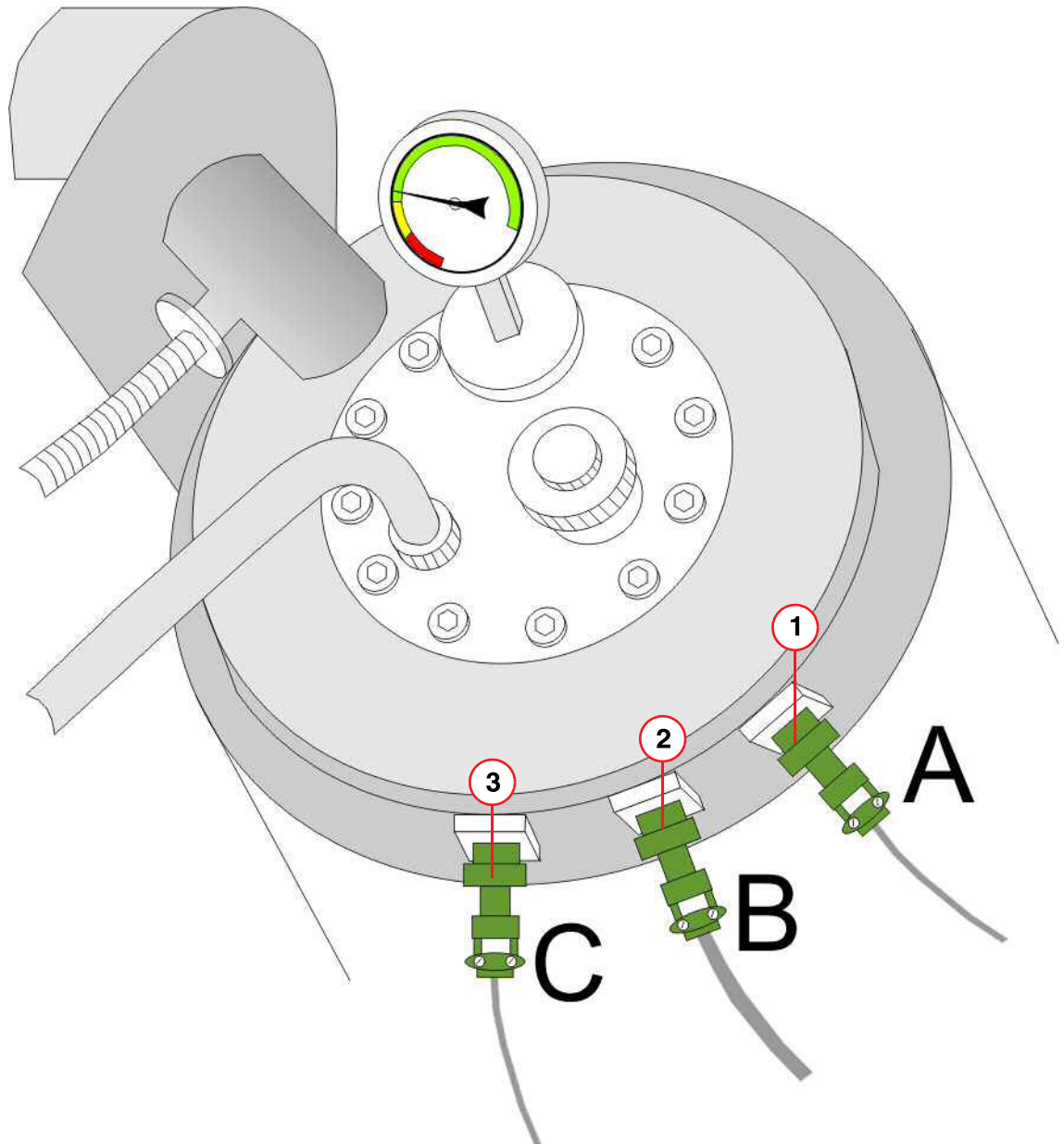
The Venting System consists of:

- **Vent Valve (1)** - A 2/3-psig (g = gauge pressure or relative pressure) non-return valve keeps the magnet He pressure at 2/3-psig. On newer systems, a single 0.5-psig valve is fitted.
- **Depressurization Valve (not visible, close to Vent Valve)** - A hand-operated bypass valve is fitted in parallel to the vent valve to allow depressurization of the helium vessel.

- **Quench Valve (2)** - The quench valve is realized as a spring-loaded clap valve which opens at 15-psig (g = gauged: pressure difference). In addition, the clap valve has a graphite burst disk rated at 20 psig which will burst if the valve should not open. Under normal circumstances, the burst disk will not break during a quench. Nevertheless, the complete valve assembly must be checked for integrity afterward.
- **Quench Tube (3)** - Exhaust line for boiled-off helium. The line connects the magnet venting system to atmosphere outside the building.
- **Pressure Gauge (4)** - To check the internal magnet pressure.
- **Helium Fill Syphon (5)** - A vacuum insulated line to fill the magnet efficiently with liquid helium (= High Efficiency Syphon or "HES"). The syphon line is routed to the side of the magnet for easy access. Older systems are equipped with a different syphon type that contains an automatic helium fill valve (= Smart Syphon).
- **Service Port for Current Lead (6)** - To energize the magnet, a so-called "Enhanced Current Lead" (ECL is part of the MPS3600 Standard Extension Kit) must be inserted into this port to connect the magnet coil to the Magnet Power Supply.

3.3.2.2 Electrical interfaces to the Magnet Supervision (MSUP)

Fig. 6: Electrical interfaces to MSUP



- (1) Port A, helium level probe
- (2) Port B, quench and switch heaters, screen reset
- (3) Port C, carbon resistors

The MSUP, located at the side of the magnet, is connected via an adapter box to Port A, B and C of the Service Turret and to Port D (Port D = located at the side of the magnet = cryo shield temperature measurement diodes).



On older OR70 magnets, the turret connectors are mounted in a different position.

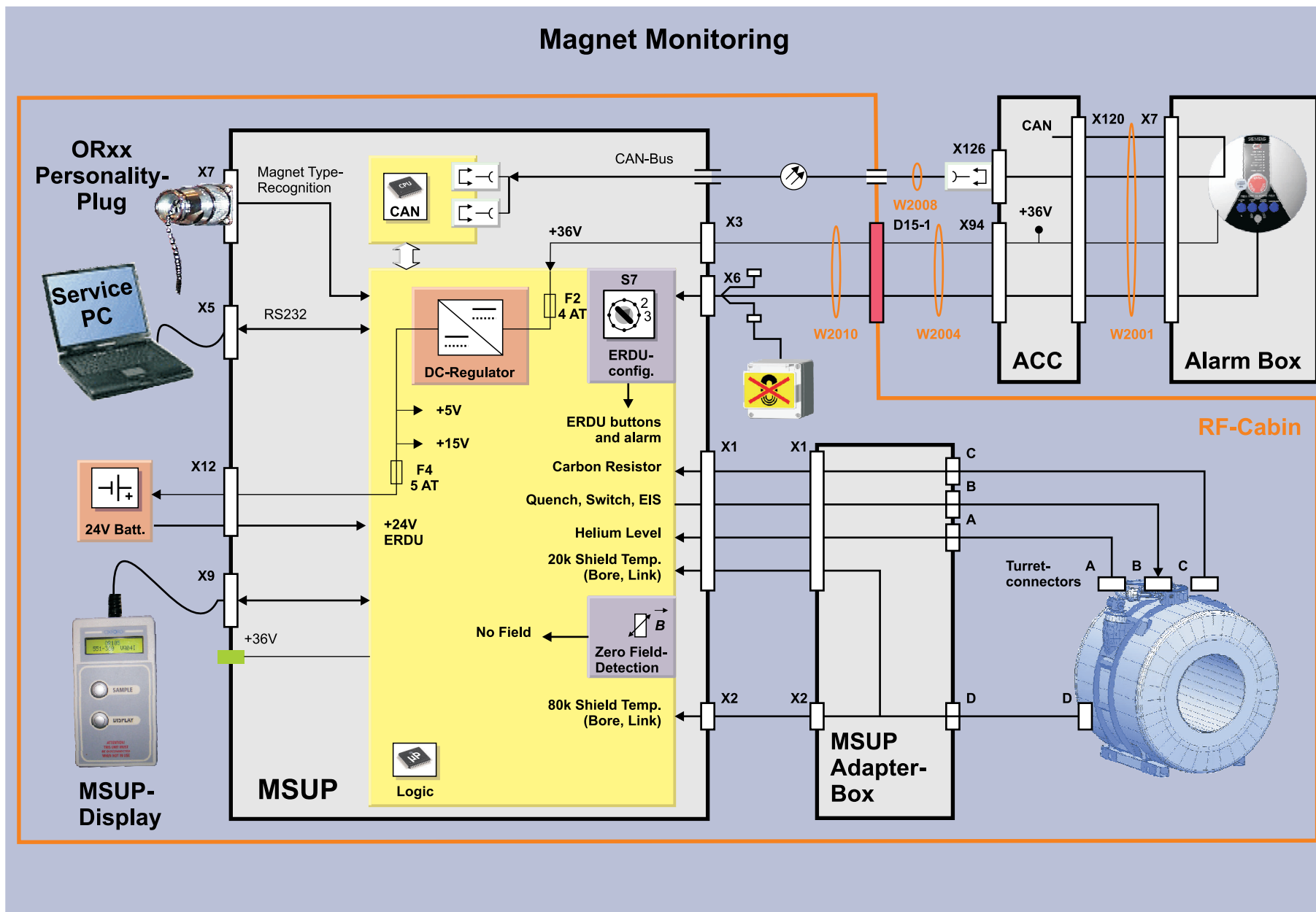
4.1 Overview

The Magnet Monitoring supervises and controls magnet functions constantly during normal operation as well as operation of the Emergency Run Down Unit (ERDU).

The Magnet Monitoring consists of:

- **Magnet Supervision (MSUP)**
- **MSUP Adapter Box** - to adapt the MSUP to the OR70 electrical interfaces
- **MSUP Battery** - to backup ERDU and essential MSUP functions
- **MSUP Display** - a service tool to perform manual readings

Fig. 7: Magnet Monitoring

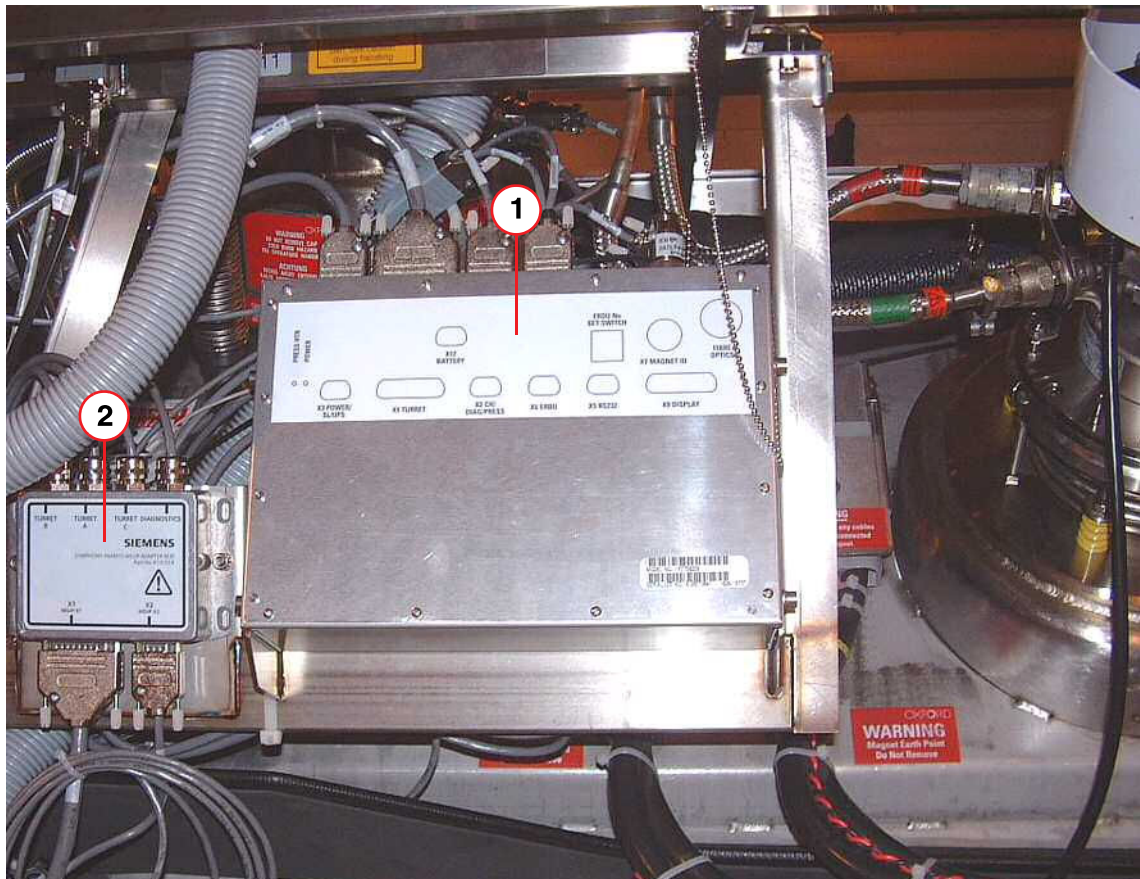


4.2 MSUP

4.2.1 General

Symphony Tim Upgrade Systems are equipped with a MSUP that is mounted at the magnet side, very similar to Avanto/Espree systems. The presence of the 36 VDC supply voltage from the ACC is indicated by a green POWER ON LED at the top of the MSUP box.

Fig. 8: Magnet Supervision (MSUP)



- (1) MSUP
- (2) MSUP Adapter Box

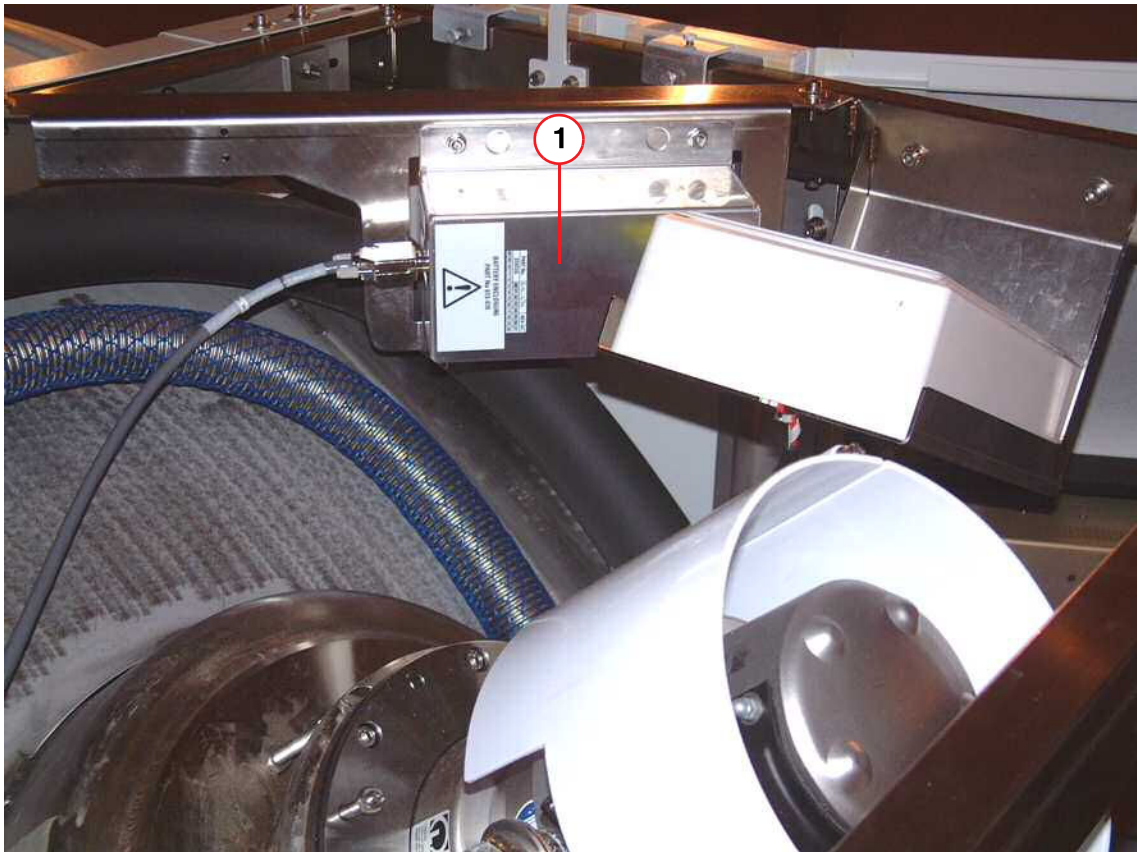
4.2.1.1 MSUP Adapter Box

The MSUP Adapter Box adapts the two MSUP connectors X1 and X2 to the four magnet connectors Port A-D.

4.2.1.2 MSUP Battery

An external 24 V rechargeable battery backs up the ERDU function. It also provides power to enable manually-triggered sensor readings such as the LHe level that can be viewed on the handheld MSUP display. The battery is located on top of the magnet frame, close to the intercom loudspeaker. It has to be replaced every two years during maintenance.

Fig. 9: MSUP Battery



(1) MSUP battery housing

4.2.2 Function

The MSUP contains the following function blocks:

- Communication Interfaces to CAN bus, Service PC and MSUP Display
- Internal Self Test and Magnet Type Recognition
- Emergency Run Down Unit (ERDU)
- MSUP Battery
- Switch and Quench Heater Control
- External Interference Screen (EIS) control
- Helium Level Monitoring
- Magnet Temperature Monitoring
- Zero Field Detection

4.2.2.1 Sleep Mode

Some of the above mentioned functions would interfere with the MR signal when performed during measurements. Therefore, the MSUP is set into a "sleep mode" during measurements. During this time, only essential functions remain active:

- ERDU circuit

- Zero Field Detection
- Battery-charging circuit

The MSUP will recover from "sleep mode" and return to "normal mode" if:

- The host transmits a wake-up CAN command
- An ERDU button is pressed
- System off is pressed
- Zero field is detected (quench)

4.2.2.2 Communication Interfaces

CAN bus

The MSUP communicates to the AMC via a fiber optic CAN bus. Changes of the MSUP status and error messages are reported via CAN commands and MPCU to the host. In the other direction, the host can load parameters to the MSUP, e.g. loadware or calibration parameters.

RS232 Serial Interface (for Service PC)

For service purpose (e.g. logfile analysis with Headquarters support), a PC can be connected to X5 of the MSUP via the HyperTerminal. Communication parameters to be set are:

- 9600 Baud rate
- 8 bit, 1 start bit, no parity
- Flow Control = None
- File, properties, settings tab, emulation = VT100.

After pressing "Esc" and typing "R" at the Service PC, the communication link is enabled.



In sleep mode, communication to the MSUP via the Service PC does not work. To wake up the MSUP, a button of the patient table control panel can be pressed.

MSUP Display

A handheld display unit (Service Tool) can be connected to the MSUP to display sensor values (e.g. LHe level, shield temperatures) and trigger actions (see MSUP display description).

4.2.2.3 Power-up Self Test

During power up, the processor initializes all MSUP components and performs a self-test during which the presence of the +5 V, +15 V and +36 V supply voltage is checked. In addition, some reference voltages are checked to verify the proper function of the multiplexer and ADC circuit for the analog sensors (e.g. LHe level measurement).

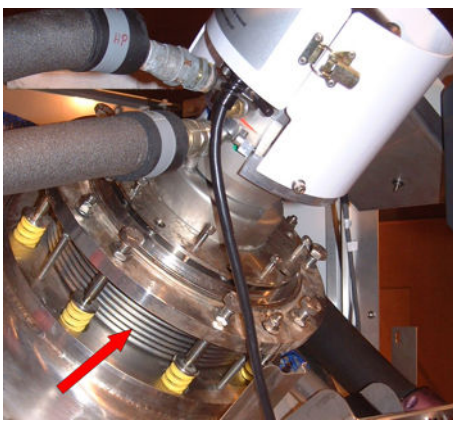
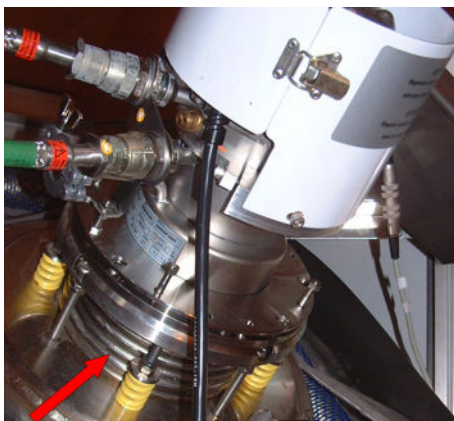
4.2.2.4 Magnet Type Recognition (Personality Plug)

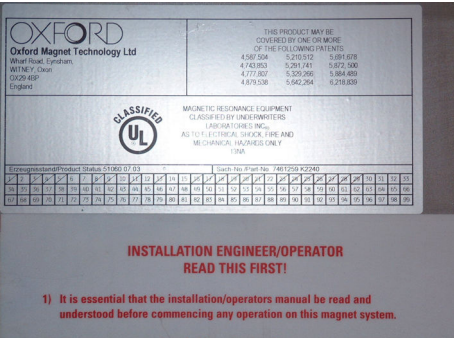
After the power-up self test, the processor extracts the Magnet Recognition from the Personality Plug that is connected to X7 of the MSUP. This is necessary to apply the correct magnet-specific monitoring parameters, warning or alarm thresholds, and to set up a connected Magnet Power Supply accordingly.



Two different types of OR 70 magnets exist, therefore two different Personality Plugs for plug code 0701 or 0070 are available.

Tab. 2 OR70 Magnet types

	OR70 with UNISOCK Mark I	OR70 with UNISOCK Mark II
UNISOCK View		

	OR70 with UNISOCK Mark I	OR70 with UNISOCK Mark II
Magnet Revision Level (Label near service platform)	Rev. Level < 25 	Rev. Level ≥ 25 
Plug Code at MSUP Display		

4.2.2.5 Supervisory Test Cycle

Automatic Test

Usually, the MSUP timer automatically starts a Supervisory Test Cycle at night to perform:

- Helium level measurement
- Battery test (only if +36 V MSUP supply voltage is present)
- EIS reset

The test cycle is completed within 30 seconds. Since this test should be performed only when the system is not used for imaging, a typical time setup in syngo "Magnet&Cooling Initialization" will be 2:00 am. The actual time information is automatically uploaded to the MSUP.

Manual Test

Additional Supervisory Tests can be invoked manually by:

- Pressing the "SAMPLE" button at the MSUP Display
- CAN bus commands (e.g. during reboot scanner)

4.2.2.6 EEPROM

The MSUP contains an EEPROM to store a lot of useful information for service. Most of the information is read by the host computer via the CAN bus and can be seen under **C:\Med-Com\log\MsupHistory.log**.

For advanced troubleshooting, the EEPROM can be read out with help of a Service PC that is connected to the MSUP RS232 serial interface (Headquarters Support required). The following information is available:

Storage of status entries

Status entries are stored with a time and date stamp after each Supervisory Test. A record of the last 256 status entries is kept.

Storage of events

The following events are stored with a time and date stamp and can be read out with the Service PC connected to the MSUP:

- ERDU button pressed
- ERDU button released
- Zero Field Detection
- Compressor start time and status
- Compressor stop time and status
- Most recently recorded helium level readings at supervisory test
- Most recently recorded shield temperatures at supervisory test
- Most recently recorded magnet pressures at supervisory test

Storage of user-configured MSUP parameters

Some of the MSUP parameters can be configured by the user (e.g. calibration of both LHe level meters in SESO) and will be stored in the MSUP EEPROM:

- He zero percent calibration value
- He hundred percent calibration value
- He level warn point
- Magnet operating current (retrieved by CAN bus from the entered MPS current value)
- Supervisory test time

Validity of the above data is checked permanently by checksum calculation. If a deviation is found, the processor tries to load the internal default parameter table for the recognized magnet type OR70.

4.2.2.7 Emergency Run Down Unit (ERDU)

ERDU buttons

The system contains two ERDU buttons (alarm box and magnet room), a third button can be fitted optionally. When a button is pressed in an emergency situation, the MSUP energizes the two quench heaters and the switch heater. The magnetic field will decay in less than 20 seconds.



The number of connected ERDU buttons has to be configured with a rotary switch S7 inside the MSUP (switch is located behind the cover close to connector X5, standard setting is "2").

ERDU button monitoring

A resistor, connected in parallel to each ERDU button, allows the monitoring of the circuitry. The MSUP drives a small constant current that generates a proportional voltage drop on each of the ERDU button resistors. In case a cable is broken or shorted, the MSUP will generate a fault message and a red LED (MAG STOP) at the Alarm Box will light up. When the ERDU has been activated the time/date of its operation is recorded in the EEPROM. The same is true for its release.

4.2.2.8 MSUP battery

The MSUP is equipped with an external 24 V battery to back-up the following vital functions:

- **Backup of Emergency Run Down Unit (ERDU)**
- **Backup of important MSUP functions** - except EIS reset

The battery lasts for 25 days after power failure (assuming one LHe measurement per day and battery was fully charged by the MSUP). The actual voltage can be displayed at the MSUP Display.



The MSUP battery must be replaced every 2 years during preventive maintenance.

ERDU battery test

The condition of the ERDU battery is tested when the MSUP starts a Supervisory Test Cycle (e.g. via timer control at night). The MSUP supply voltage (+36 VDC) must be present for that test. A load resistor is connected across the battery and a comparator checks that the battery voltage is within the specified limits. If the battery voltage is less than +23.25 V, a warning message is generated although scanning is still possible. If the battery voltage is less than +23 V, an alarm message is generated and all scanning is blocked. The red "MAG STOP" LED lights up (in either case) on the alarm box.

4.2.2.9 Switch and Quench Heater control

The Switch Heater and Quench Heaters of the magnet are controlled and constantly monitored by the MSUP. A small constant current is driven through all heater circuits and the heater resistance values are measured to check for broken or shorted lines. An alarm is generated if a heater resistance value is found to be outside specification.

4.2.2.10 Helium Level monitoring

The helium level inside the magnet can be measured with the help of two independent superconducting probes built into the helium vessel.

When the sampling is started, the MSUP performs an auto-calibration of the LHe measurement current that is applied to the probe. The current is slowly increased from zero until the probe starts propagating (i.e. the parts of the probe inside the gaseous helium heat up until they change to the normal conducting state, parts in the liquid stay superconductive). Then, the current is reduced to 250 mA with a voltage drop across the probe measured and converted into a LHe filling percentage.

Display of last sampled LHe level

The LHe filling level of the magnet can be seen at:

- syngo MR, System / Control
- Service Software (SESO), Magnet&Cooling Status
- MSUP display

LHe level sampling

A new sampling of the LHe level can be started:

- Automatically one time per day via the internal MSUP timer (sample time setup in SESO, Magnet&Cooling Initialization)
- Manually by calling SESO, Magnet&Cooling Status
- Manually at the MSUP Display

Calibration of LHe level meters

Different probes may vary in resistance, therefore the resistance values for 0% and 100% He-level (to be found in the SMT Magnet Acceptance Certificate) need to be entered under SESO Magnet&Cooling Initialization.

Helium Level warning and alarm

A warning level (possible range 40-79%) for LHe level can be adjusted in SESO "Magnet&Cooling Alarm Settings". If the level falls below that threshold, a warning message is generated and reported to syngo MR.

The alarm threshold is fixed at 35%. If the level falls below that value, the red "Helium Level" LED lights up at the Alarm Box and the alarm buzzer is switched on.

4.2.2.11 External Interference Screen (EIS) control

To eliminate any accumulated current in the External Interference Screen (EIS) coil windings that may deteriorate the B_0 field homogeneity, the EIS coil has to be quenched once a day. The MSUP internal timer switches on the EIS heater for 15 seconds, the coil gets normal conductive and the current is released. To avoid image quality problems, the EIS has to be reset when the system is not utilized (typical time set-up = 2:00 am).

A manual EIS reset can be performed with help of the MSUP Display (e.g. during shimming). If forgotten, the MSUP will switch off the EIS heater automatically after 60 minutes to avoid excessive helium boil-off.

The presence of the EIS reset output current is checked by the MSUP circuitry. In case of a fault, (e.g. broken cable), a warning message is generated and the yellow "EIS"-LED at the Alarm Box lights up.

4.2.2.12 Magnet Temperature Monitoring

The following magnet temperatures can be checked by the MSUP:

- 20 K shield temperature - link and bore temperature
- 80 K shield temperature - link and bore temperature
- Carbon resistors - 4x

Temperature measurements are achieved by passing a constant current through the sensors/resistors and monitoring the voltage readings. They are partially converted into temperature values.

4.2.2.13 Zero-Field detection

The MSUP contains a sensor to detect the absence of the main magnetic field (= Quench detection).

4.2.3 Specifications

Tab. 3 MSUP Parameter Specifications

Parameter	OR70 Rev. Level < 25 (UNISOCK Mark I)	OR70 Rev. Level ≥ 25 (UNISOCK Mark II)
He sample pulse height	0.25 A	
Number of daily samples per day	1	
He level alarm set	35 %	
He level warning set	40 - 79 %	
Check on battery voltage warning level	23.25 V	
20 K Bore Shield temp warn.	> 101 K	> 86 K
20 K Bore Shield temp alarm	> 106 K	> 91 K
20 K Link Shield temp warn.	> 91 K	> 75 K

Parameter	OR70 Rev. Level < 25 (UNISOCK Mark I)	OR70 Rev. Level ≥ 25 (UNISOCK Mark II)
20 K Link Shield temp alarm	> 96 K	> 80 K
80 K Bore Shield temp warn.	> 101 K	> 86 K
80 K Bore Shield temp alarm	> 106 K	> 91 K
80 K Link Shield temp warn.	> 91 K	> 75 K
80 K Link Shield temp alarm	> 96 K	> 80 K
Maximum EIS ON time-out	60 min.	

4.2.4 Switches

Tab. 4 Switches, MSUP

Switch	Location	Description
S7	Inside MSUP	Configuration: number of ERDU buttons

4.2.5 Connectors

Tab. 5 MSUP connectors

Connector	Location	Connection to
X1	MSUP	X1 MSUP Adapter Box
X2		X2 MSUP Adapter Box
X3		ACC, MSUP Power +36 V
X4		Not Used
X5		RS232 Service PC
X6		ERDU buttons
X7		Magnet Type Recognition, ORxx Personality-Plug
X8		Not Used
X9		MSUP display
X12		MSUP battery
CAN Optical Waveguide		Fiber optic CAN bus to X126, ACC

Tab. 6 MSUP Adapter Box connectors

Connector	Location	Connection to
X1	MSUP Adapter Box	X1 MSUP
X2		X2 MSUP
Plug A		Magnet Port A (LHe level probes)
Plug B		Magnet Port B (Switch/Quench/EIS-Heaters)
Plug C		Magnet Port C (Carbon Resistors)
Plug D		Magnet Port D (Shield Temperature Diodes)

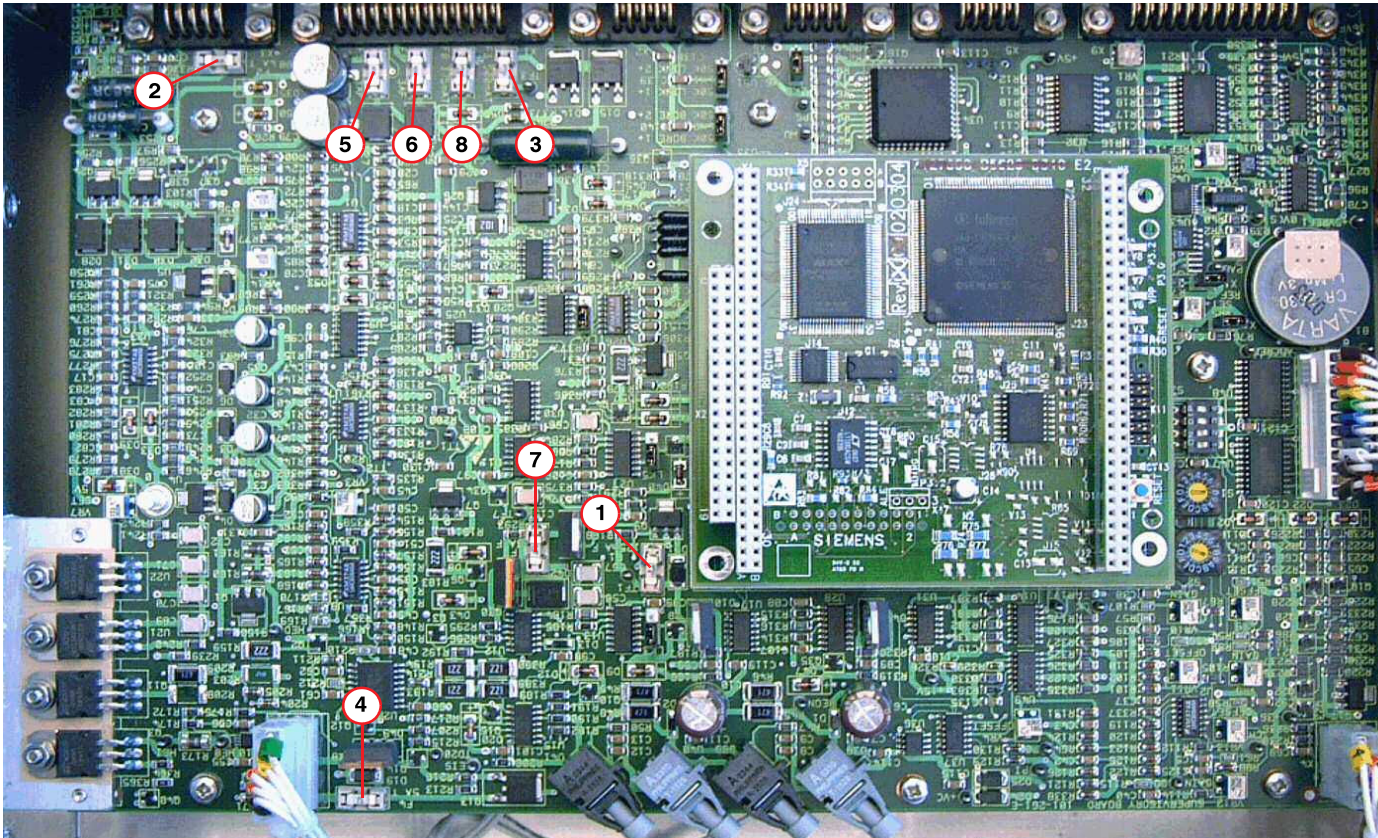
4.2.6 LEDs

Tab. 7 LEDs, MSUP

LED	Location	Description
Green	Top of MSUP box	+36 V MSUP supply voltage present

4.2.7 Fuses

Fig. 10: MSUP Fuses



- (1) Fuse F1, 125 mA, EIS Control (250 mA for MSUP with Rev. ≥ 14)
- (2) Fuse F2, 4 AT, +36 V MSUP supply
- (3) Fuse F3, 500 mA, LHe level probe
- (4) Fuse F4, 5 AT, Battery Supply
- (5) Fuse F5, 3 AF, Quench Heater 1
- (6) Fuse F5, 3 AF, Quench Heater 2
- (7) Fuse F7, 3 AF, Pressure Heater Control
- (8) Fuse F8, 375 mA, Switch Heater (1 AF for MSUP with Rev. ≥ 14)

4.3 MSUP display

Fig. 11: MSUP Display



The MSUP Display is a Service Tool and must be disconnected during imaging to avoid interference to the MR signal.

4.3.1 General

The MSUP display is a handheld monitoring aid that can be connected to X9 of the MSUP. It provides local monitoring of the magnet parameters but also alarms and warnings if values are not within specifications. In addition, LHe level measurement and EIS-Reset can be activated by hand.

4.3.2 Function

4.3.2.1 Switching on the MSUP Display

- Connect the Display Unit to connector X9 on the MSUP
- Ensure that the +36 V MSUP supply voltage is present
- Press the "SAMPLE" button

4.3.2.2 Start a Manual Sampling

You can measure the LHe level, test the MSUP Battery Test, and reset the EIS by:

- Pressing the "SAMPLE" button

4.3.2.3 Display of magnet parameters

To display all available magnet parameters, the unit has to be unlocked first:

- Hold the "Display" button pressed and operate the "Sample" button once.
- Pressing the "Display" button repeatedly

If a value is outside its specified limits, a (W) or (A) will be displayed with the parameter reading to indicate a Warning or Alarm situation.

Tab. 8 MSUP Display parameters

Item #	Parameter	Typical Value OR 70 with UNISOCK II	Typical Value OR 70 with UNISOCK Mark I
1	He level (Press sample to start a new LHe level measurement)	35 % - 99 %	
2	Battery volts	24 - 27 V	
3	80 K link, bore (Kelvin)	52 K (link), 57 K (bore)	63 K (link), 79 K (bore)
4	Carbon Resistor 1, 2 (Ohms)	1100 - 1200 Ohms (1, 2)	
5	Carbon Resistor 3, 4 (Ohms)	1100 - 1200 Ohms (3, 4)	
6	20 K link, bore (Kelvin)	15 K (1), 16 K (2)	19 K (1), 20 K (2)
7	Switch Heater (Ohms, status)	33 Ohms, Off	
8	Quench heaters 1, 2 (Ohms)	20.8 Ohms	
9	ERDU buttons (status)	OK	
10	EIS	OFF / ON	
11	Self test	Pass	
12	Compressor status (On/Off)	ON	
-	Date and time	Date, Time	
-	Personality Plug Code, Magnet Type, Software part number and SW version	e.g. 0070 OR70-2 551-360 VA06H	e.g. 0701 OR70-2 551-360 VA06H

4.3.2.4 Perform a manual EIS reset

For service purposes (e.g. during shimming), the EIS can be manually activated:

- Scroll to parameter "EIS" by pressing the "Display" button repeatedly
- Press the "Sample" button to toggle the EIS on or off

4.3.2.5 Switching OFF MSUP display

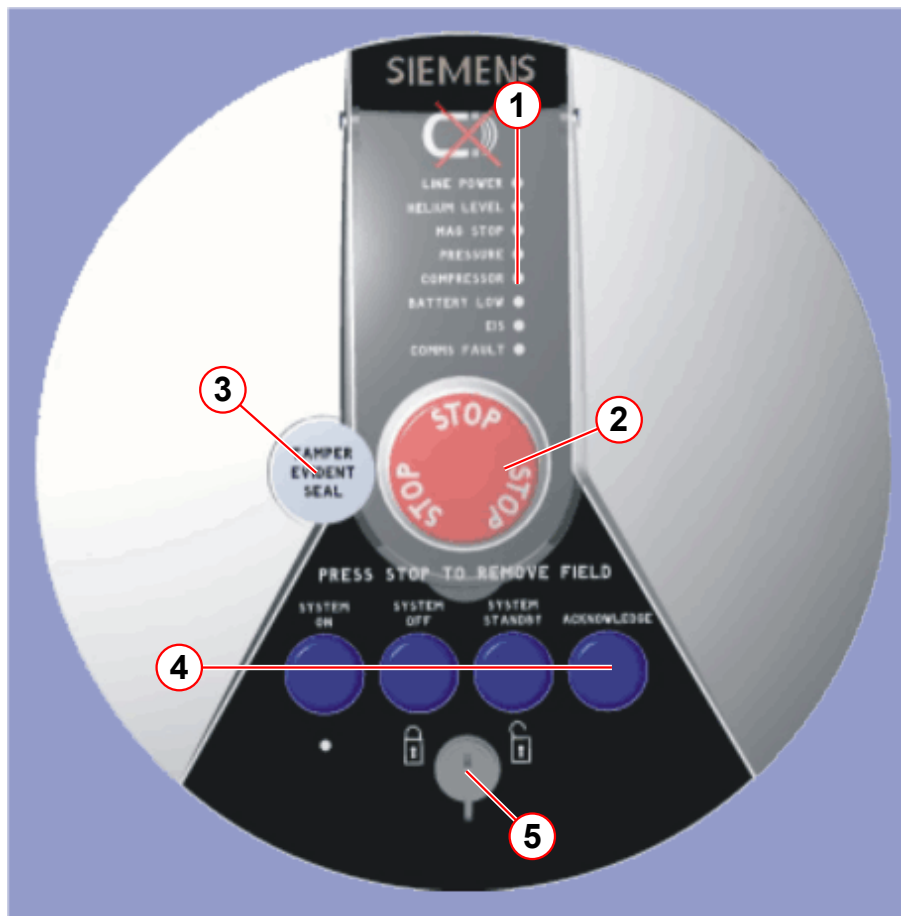
Disconnect the unit's connector X9 on the MSUP.



If the +36 V MSUP supply voltage is not present, the MSUP display will switch off automatically after 2 minutes.

5.1 Overview

Fig. 12: Alarm Box Front Panel



- (1) Status and error LEDs
- (2) ERDU button
- (3) Tamper proof seal
- (4) System power switches and alarm acknowledge
- (5) Power switch locking key

The Alarm Box is a user interface for:

- **Power switching of the MR-system**
- **Magnet Supervision Indicators**
- **Remote control of He-Compressor**
- **ERDU**

5.2 System power switching

The **System ON** and **System OFF** buttons are hard-wired to relays k10 and k11 in the Power Distributor responsible for these functions.

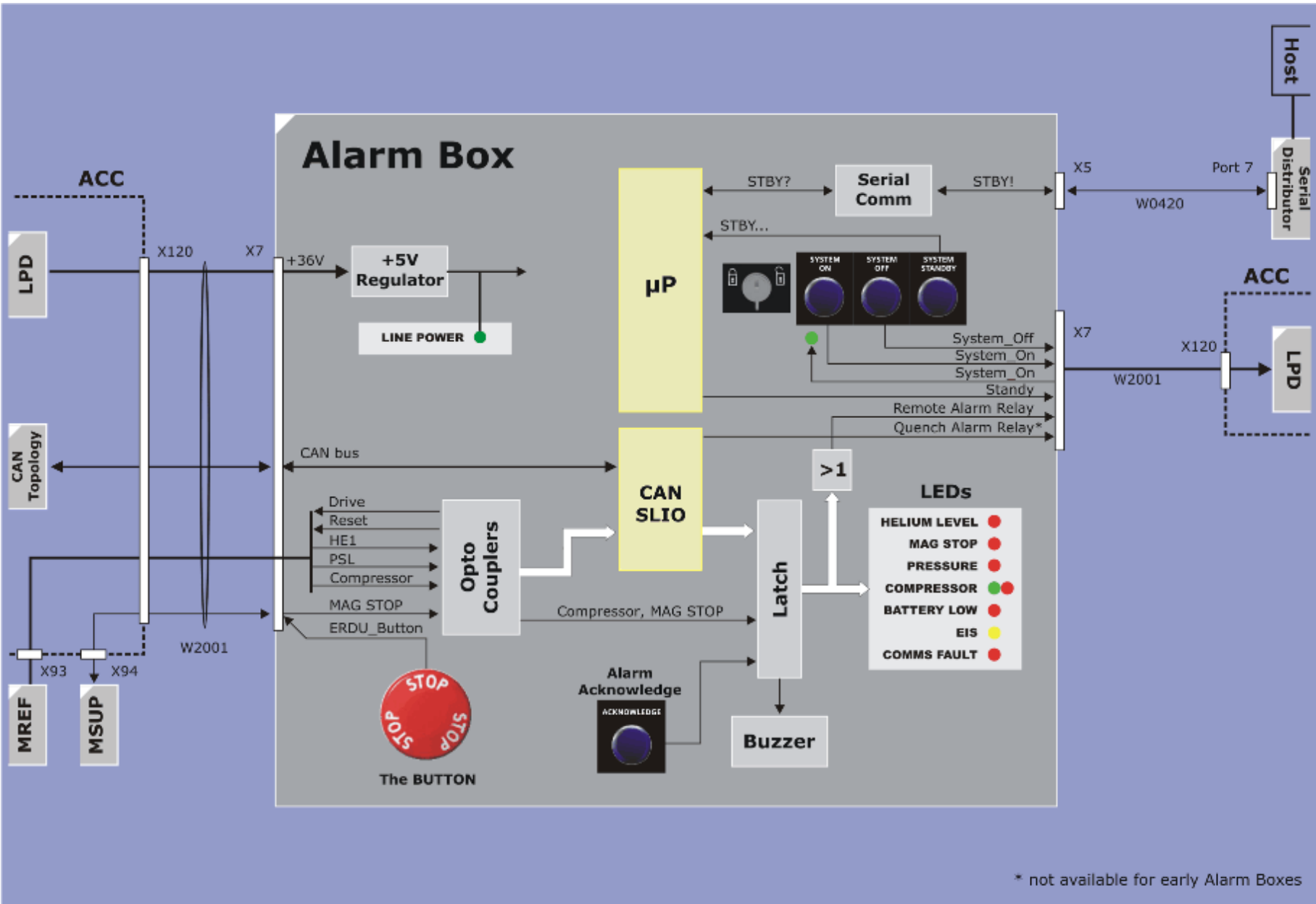
When the **System STANDBY** button is pressed, a command is sent to the Host via a serial communication line. The Host in turn shuts the Imager down, after which the Host sends back a command to the Alarm Box over the same serial line to activate the System Standby signal to the PDS (relay k13).

5.2.1 Buttons

Tab. 9 Alarm Box switches

Switch	Description
SYSTEM ON	All MR-System components are powered up
SYSTEM OFF	All MR-System components are switched off
SYSTEM STANDBY	All components except the host are switched off
LOCKING KEY	To prevent from unauthorized power switching
STOP BUTTON	Emergency Run Down Unit (ERDU) for magnet
ACKNOWLEDGE	Resets error latch and buzzer

Fig. 13: Alarm Box Block Diagram



5.3 Status Indicators

There are several Alarm Box status indicators which can trigger an external **Customer Alarm** contact:

- HELIUM LEVEL too low
- BATTERY LOW
- COMPRESSOR OFF
- MAGNET PRESSURE too low (not used for Symphony / Trio A Tim System)
- SHIELD TEMPERATURE too high

In case of alarm, the Alarm Box activates relay K17 in the Line Power Distribution system. Galvanically isolated contacts are supplied to the customer over connector X119 at ACC roof. The customer alarm can be reset by pressing the "ACKNOWLEDGE" button.

New Alarm Boxes additionally provide a so-called **Dedicated Quench Alarm**.

This alarm is triggered when the MSUP reports "no field" and "magnet pressure too high" simultaneously. The alarm is fed to an additional relay K19 in the ACC Power Distribution System which drives a potential-free contact to be accessed at connector X119 ACC top (see also description "Power Distribution").



At Trio A Tim Systems it is strongly recommended to utilize the Customer Alarm to detect the critical LHe level after an unattended quench. The magnet will warm up if not refilled within 6-10 hours!

5.3.1 LEDs

Tab. 10 Alarm Box LEDs

LED	Description
LINE POWER	Power is applied to the Alarm Box
HELIUM LEVEL	Helium level alarm < 40 %
MAG STOP	Quench / Switch Heater fault or ERDU button / battery fault
PRESSURE	Magnet pressure too high / low (not used at Symphony- / Trio A Tim-Systems)
COMPRESSOR	Green = compressor running Red = compressor has stopped by fault
BATTERY LOW	ERDU battery voltage < 23.25 V

LED	Description
EIS	Fault in the External Interference Screen circuit
COMMS	To ensure the integrity of the communication link, the MSUP triggers a watchdog at the Alarm Box. When communication is down for more than 10 seconds, the watchdog sets the "COMMS" fault.

5.3.2 Connectors

Tab. 11 Alarm Box connectors

Connector	Location	Description
X5	Alarm Box	Remote power switch from host serial port #7
X7	Alarm Box	Power switching signals, CAN0-bus connection, He-compressor control
X119	ACC roof	External customer alarm (pin 1 = N/O, pin 2 = N/C, pin 3 = common). External quench alarm (available on newer systems only; pin 4,5 = N/C). For details please refer to the Functional description of "Power Distribution" and to the Planning Guide.

5.3.3 Jumpers

Tab. 12 Alarm Box jumpers

Jumper	Position	Description
LK1	pin1-pin2	Alarm buzzer normal operation
	pin2-pin3	Alarm buzzer disabled (not recommended!)

5.4 Compressor Control

The Alarm Box is connected via connector X7 to X120 at ACC roof. Various compressor related signals are routed via that cable harness.

A connection via X93 ACC to the EXT. JR connector of the compressor connects the discrete signal lines for:

- Compressor status and error signals
- Reset signal to compressor
(after power-, water- and temperature-failures)
- Compressor switch-off signal
(in case pressure control detects a low magnet pressure)

Compressor related status and error signals are communicated via CAN0-bus connection to the MSUP and host.

See Refrigerator section for wiring diagram of the compressor remote control.

5.5 ERDU Stop Button

Two functional parts of the Emergency Run Down Unit (ERDU) are located at the Alarm Box:

- Red ERDU "STOP" push button
- Red "MAG STOP" LED

5.5.1 Function

When the ERDU button is pressed in case of emergency, the MSUP energizes the quench and Switch Heaters of the magnet, resulting in a quench of the superconducting magnet coil. The magnetic field decays immediately. To protect against accidental operation, the button cover is protected by a seal. The seal must be replaced after operation of the button.

6.1 Overview

With the help of the Advanced High Order Shim, the local magnetic field inhomogeneities caused by the Patient can be improved.

Passive shimming with iron plates reduces low and high order inhomogeneities caused by magnet tolerances and ferromagnetic components in the environment. Patients (and phantoms) also cause local field inhomogeneities that have to be compensated for applications which are sensitive to inferior shim, e.g. Spectroscopy, Fatsat, EPI (Echo Planar Imaging), TGSE (Turbo Gradient Spin Echo).

The correction of linear inhomogeneities is done via offset currents applied to the Gradient Coils. Second order inhomogeneities are compensated by applying currents to 5 Shim Coils that are integrated into the Gradient Coil:

- $A(2,0)$
- $A(2,1)$
- $B(2,1)$
- $A(2,2)$
- $B(2,2)$.

(see also Tune-up, Phantom Shim)

On systems where the Advanced High Order Shim is an option, the following components need to be installed separately:

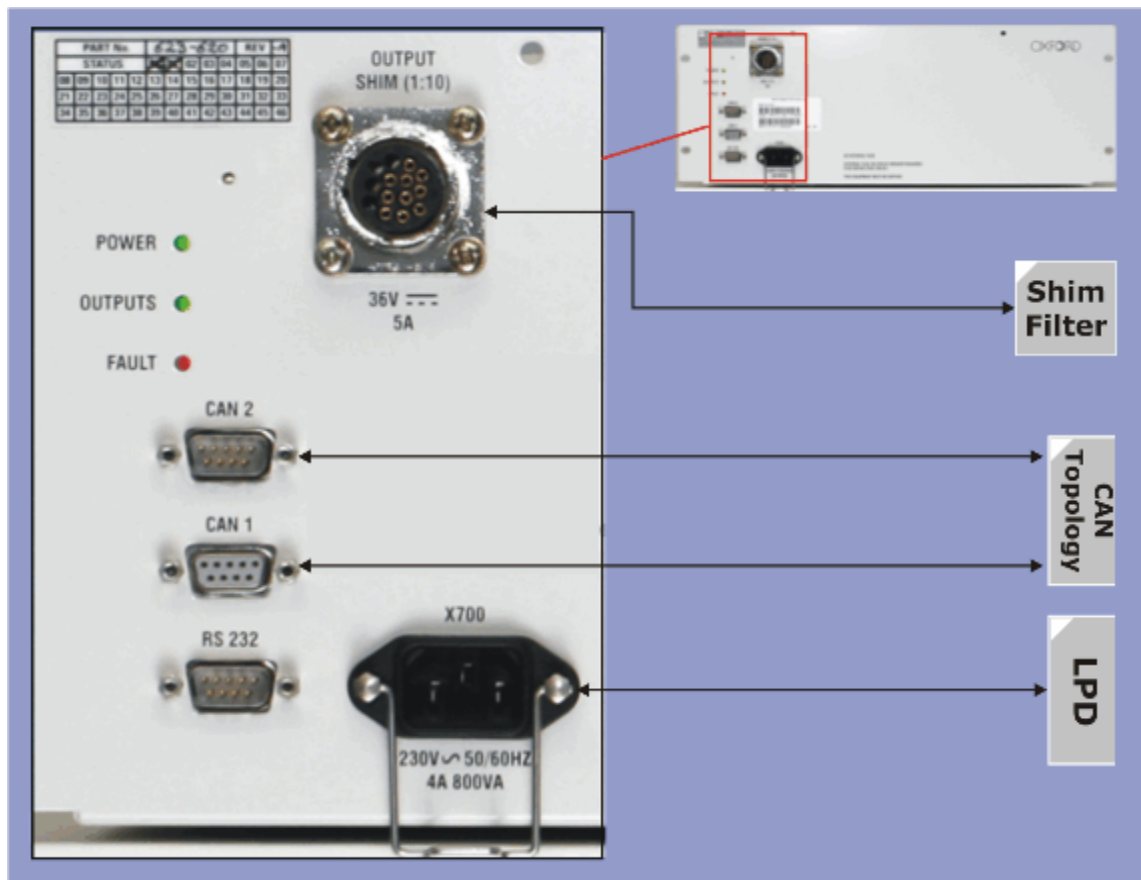
- **Shim Power Supply Unit (SPSU)**
- **Shim Filter Unit**
- **Shim cables**

The SPSU is mounted inside the ACC cabinet, the Shim Filter Unit is installed on the filter panel.

After installation of the hardware, the Advanced High Order Shim has to be enabled in the syngo MR configuration.

6.2 Shim Power Supply Unit (SPSU)

Fig. 14: Shim Power Supply Unit



6.2.1 General

The SPSU is contained in a metal box and mounted inside the ACC cabinet. It comprises one large printed circuit board containing most of the electronic parts and a small display board for front panel connectors and LEDs. The base of the unit contains two switched-mode power supplies and a cooling fan, together with a wiring loom and structural metal components.

6.2.2 Function

Five built-in switch-mode DC/DC converters are responsible for producing the shim currents for the 5 Shim Coils. The maximum current that can be supplied is 5 A per channel at a maximum output voltage of +/- 20 V. Current regulation is based on the principle of Pulse Width Modulation.

Shim currents are setup under syngo MR in the Tune-Up step "Phantom Shim". They are loaded automatically into the SPSU via its CAN-bus connection. If required, these shim currents can be optimized temporarily for a patient measurement with the help of the 3D-Shim. The SPSU is able to report its status via CAN commands to syngo MR, i.e. error messages can be seen in the eventlog.

6.2.3 Specifications

Tab. 13 Shim Power Supply Unit specifications

Shim Power Supply Unit 2362	Specification
Line voltage:	198-253 VAC, 50/60 $\pm$ 1 Hz
Input power:	< 0.8 kVA
Maximum current per channel:	$\pm$ 5 A DC, simultaneously on all channels
Maximum voltage per channel:	$\pm$ 20 VDC
Shim Coil resistance A(2,0):	< 1.5 Ohms
All other coils:	< 2.0 Ohms

6.2.4 Connectors

Tab. 14 Shim Current output connector

Shim PSU-channel#	Shim Coil	Shim filter-Pin#	Shim connector at Gradient Coil; Pin#
Channel 1 +	A20 +	1	14
Channel 1 -	A20 -	2	24
Channel 2 +	A21 +	3	1
Channel 2 -	A21 -	4	3
Channel 3 +	B21 +	5	10
Channel 3 -	B21 -	6	12
Channel 4 +	A22 +	7	26
Channel 4 -	A22 -	8	28
Channel 5 +	B22 +	9	35
Channel 5 -	B22 -	10	37

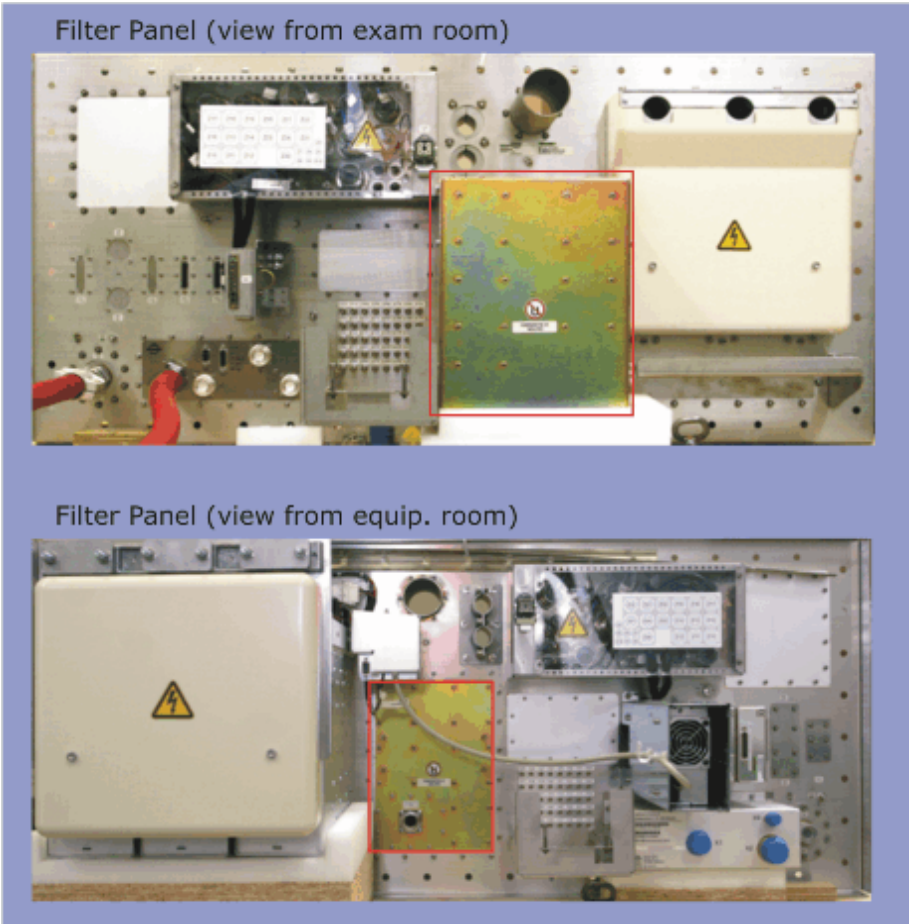
6.2.5 LEDs

Tab. 15 Shim Power Supply LEDs

LED	Description
POWER, green	If lit, SPSU is in standby
OUTPUTS, green	Shim Outputs are switched on
FAULT, red	Error has occurred

6.3 Shim Filter Unit

Fig. 15: Shim Filter Unit



6.3.1 Function

To supply the Shim Coils of the scanner inside the examination room, the output leads of the SPSU are fed through an RF filter box mounted at the filter plate. By that means, external interferences riding on the outside shim cables can not pass into the RF-cabin. The filters also protect the SPSU from voltage transients induced by the Gradient Coil.

6.3.2 Specifications

Tab. 16 Shim Filter Unit specifications

Shim Filter Unit	Specification
Weight:	23 kg (50.7 lb)
Cooling:	air-cooled, convection

Shim Filter Unit	Specification
Resistance of Filter Unit plus Cables:	< 1.2 Ohm, per channel
Power dissipation:	< 88 W all outputs at 5 A

7.1 Overview

The cryogenic Refrigerator System for the OR 70 magnet consists of:

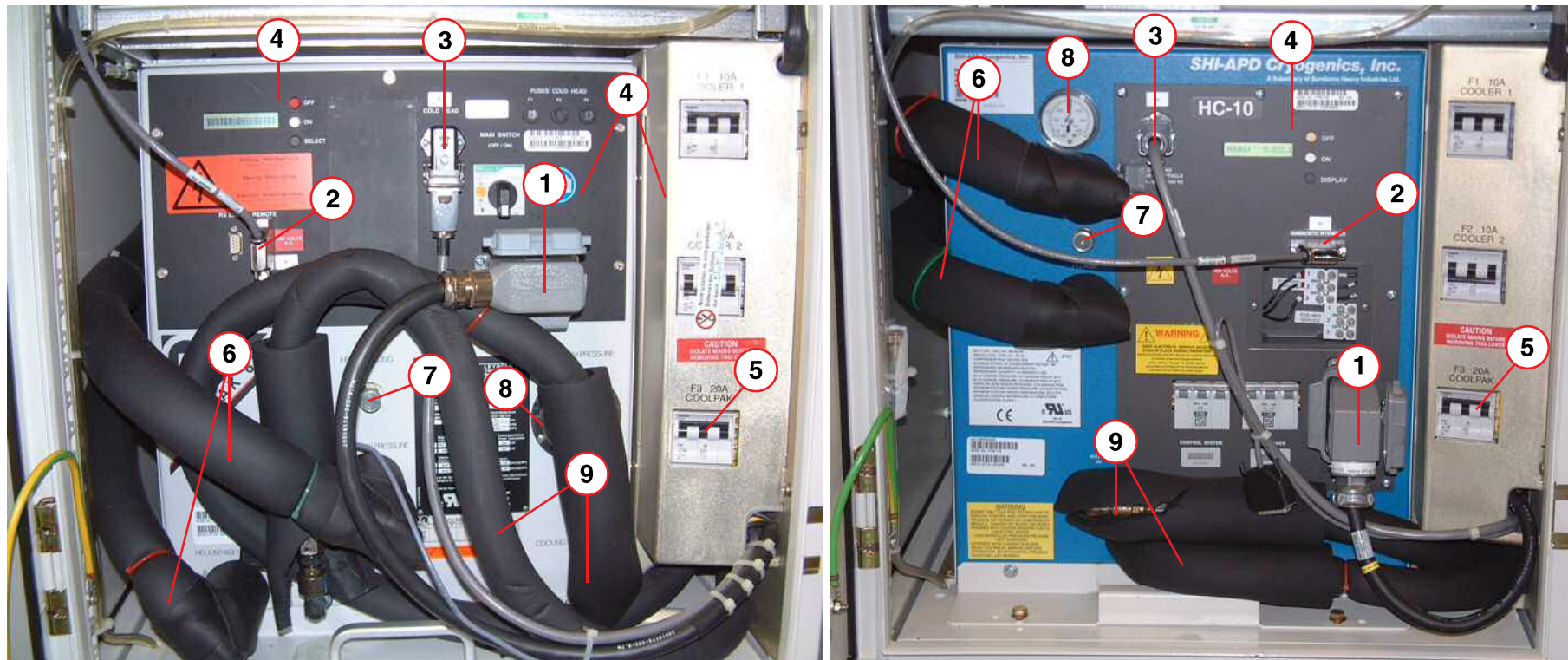
- **He-Compressor** - For the supply of high pressure helium gas for cold head operation. The compressor is water-cooled by the cooling system.
- **Cold Head** - For cooling the 20 K and 80 K magnet cryo shields in order to minimize the helium boil-off.
- **He Flexlines** - The connection between the He compressor and cold head.

The Refrigerator System is designed and delivered by LEYBOLD VACUUM GmbH, Germany. In 2001, a new refrigerator system - the SC-10/SC-10L Shield Cooler System from SHI-APD Cryogenics Inc., Japan - was introduced, first for MAGNETOM Allegra and then for MAGNETOM Sonata.

The main advantage of the SHI-APD refrigerator compared to the LEYBOLD refrigerator is that less vibrations are generated during operation, resulting in a better phase stability of the MR signal.

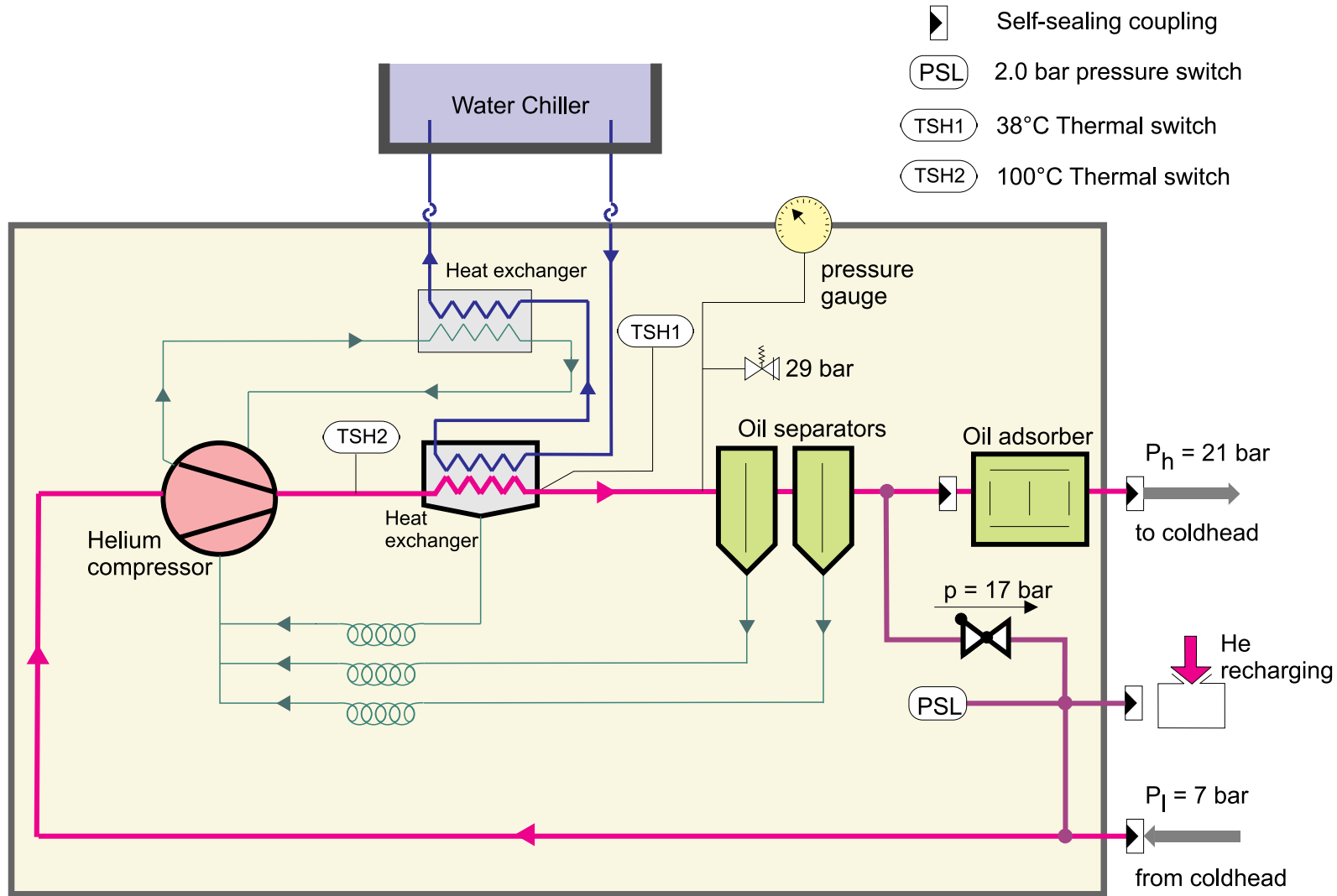
7.2 He Compressor

Fig. 16: He Compressor (left: Leybold COOLPAK 6000, right: SHI-APD HC-10)



- (1) Power Connector
- (2) Remote/Diagnostic Interface Connector X7 (Compressor status and control)
- (3) Connector to Cold Head Motor
- (4) Status and error display, operation buttons
- (5) Main Power Fuse Breaker
- (6) Helium pressure flexlines to Cold Head
- (7) Helium gas refill port
- (8) Helium pressure gauge
- (9) Water cooling

Fig. 17: Compressor schematic diagram (here: Leybold COOLPAK 6000)



7.2.1 General

The He compressor comprises the modules:

- **Compressor capsule**
- **Heat exchanger**
- **Gas purifiers**
- **Control module (electrical supply unit)**

7.2.2 Function

7.2.2.1 Compressor capsule

The oil-lubricated compressor pressurizes high purity 5.0 helium gas (purity > 99.9990 %) from approx. 7 bar (PI) to approx. 21 bar (Ph). During transportation, the compressor must not be tilted by more than 30 degrees, otherwise compressor damage or oil contamination of the cold head may occur!

7.2.2.2 Heat exchanger

After being pressurized, the hot helium gas is passed through a helium/water-counter flow heat exchanger, located downstream of the compressor. The gas temperature is lowered to about the temperature of the coolant.

A second oil/water-counter flow heat exchanger cools the oil of the compressor capsule. Oil circulation is maintained by the difference in pressure between low and high-pressure helium. The temperature of the cooling water should be 5 °C - 25 °C at a flow rate of 5...10 l/min.

7.2.2.3 Gas purifiers

Downstream from the heat exchanger, the helium is passed through a two-stage oil separator. The condensed oil from the oil separators is returned to the compressor by means of capillary pipes.

The helium gas exiting the second oil separator still contains small quantities of oil vapor. To prevent this oil from entering the cold head and freezing there, an adsorber is located behind the oil separators.



In accordance with the maintenance schedule, the adsorber must be replaced after every 18000 hours of operation (~ 2 years).

7.2.2.4 Safety Devices

The compressor unit has the following safety equipment (detail description valid for Leybold compressor only, SHI-APD compressor contains similar devices):

- **Pressure switch PSL** - shuts down the compressor unit if the pressure difference between low-pressure helium and atmospheric pressure drops below 2.0 bar. Display on the service display in case of error: 1.: SYSTEM ERROR 2.: Pressure Alarm

- **Thermal Switch TSH1** - shuts down the compressor if the temperature of the helium behind (downstream of) the heat exchanger exceeds 38° C. Display: 1.: SYSTEM ERROR 2.: Helium temp.fail
- **Thermal Switch TSH2** - shuts down the compressor if the temperature of the helium before (upstream of) the heat exchanger exceeds 100° C. Display: 1.: SYSTEM ERROR 2.: Oil cct. fail.
- **Thermal switch in compressor capsule** - Display: 1.: SYSTEM ERROR 2.: Compressor fail.
- **Motor protection switch** - Protects the compressor unit against excessive main currents. Display: Switch set to 0.
- **3 fuses for cold head (F1, F2, F3, accessible from outside)** - Display: Phase/Fuse ERROR DC Voltage error
- **2 fuses for control voltage (internal)** - Display: There will be no display or: DC Voltage error.
- **Bypass valve** - Opens a shunt between the high and low-pressure helium circuits if the pressure difference exceeds 17 bar. Display: there will be no display.
- **Safety valve** - Opens the helium circuit to the atmosphere if the high-pressure segment exceeds 29 bar.
- **Microprocessor for checking the phase rotation** - Display: Phase/Fuse ERROR.

If the compressor is shut down because of a failure, (helium pressure too low, helium temperature too high or other reasons, see the list under Safety Devices), the service display reads **SYSTEM ERROR**.

The nature of the fault can be ascertained by pressing the "Select" button.

7.2.2.5 Control module

The compressor unit is equipped with several monitoring devices to indicate faults. Compressor status and errors are reported to a service display at the front side of the compressor and/or via Remote/Diagnostic Interface connector X7 to the Alarm Box CAN controller, from which they can be accessed by the host computer and MSUP.



Two different connection cables (W2007) to the Remote/Diagnostic interface X7 exist. (SMT cable 617-372 is used on SHI-APD compressors, the 617-373 cable on Leybold units).

7.2.2.6 Power adaptation

The control module supplies the compressor capsule and the control electronics with the necessary operating voltages. It needs to be adapted to both line voltage and frequency prior to start up. The adaptation rail is located under a front cover of the He compressor.

7.2.2.7 Operating modes

Contrary to the SUMITOMO compressor used for Avanto and Espree, there is only one mode of operation. The compressor may be turned on manually at the "ON" button, or it will start anyway when the MSUP / Alarm Box sends a reset signal.

7.2.2.8 Remote control of Leybold He compressor ("Remote" plug X7)

Description pending.

(The functions are similar to the following description for the SHI-APD He compressor, nevertheless pin numbers and minor details may be different).

7.2.2.9 Remote control of SHI-APD He compressor ("Diagnostic Interface" plug X7)

There is a PCB inside the plug that connects to the SHI-APD compressor. The function of this PCB is simply to convert the signal and control lines between the compressor and the ACC, so that the Alarm Box, which is designed to control a Sumtomo compressor, can control a SHI-APD compressor in a similar fashion. The "Diagnostic Interface" plug connection must be fitted during operation of the He compressor to support:

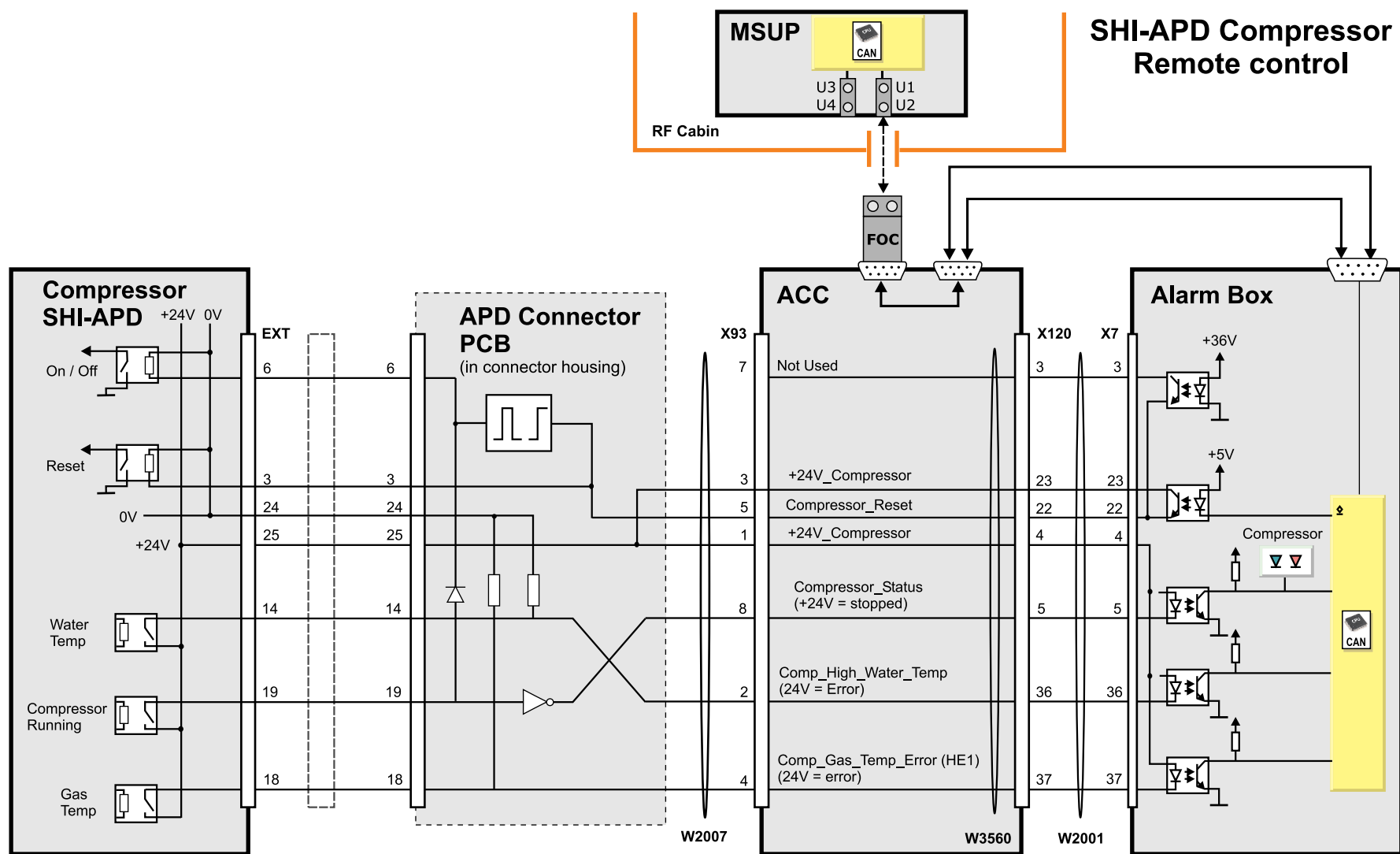
- **Status and error reporting to alarm box** - Status: Compressor running/stopped. Errors: Compressor helium gas temperature error, compressor water overtemperature.
- **Reset signal from alarm box** - The Alarm Box (re)starts the compressor by sending a "Reset" signal which the PCB in the compressor plug turns into a "Reset" followed by "Remote on" signals. The compressor is latched on by feeding back (via the PCB) its "Run Status" signal output to its "On/Off" signal.
- **Reset signal from MSUP** - This is to restart the compressor in case of power failures and internal compressor error conditions. When the compressor is sensed off, the reset signal is applied at 1 1/2 hour intervals until the compressor restarts.

7.2.3 Specifications

Tab. 17 He Compressor specifications

Compressor type	SHI-APD
Static pressure	15 - 17 bar
Dynamic pressure	19 - 22 bar
Cooling water flow	6 - 10 l/min.
Cooling water temperature	4 - 25° C
Water filter	< 100 microns

Fig. 18: Remote control of SHI-APD He Compressor



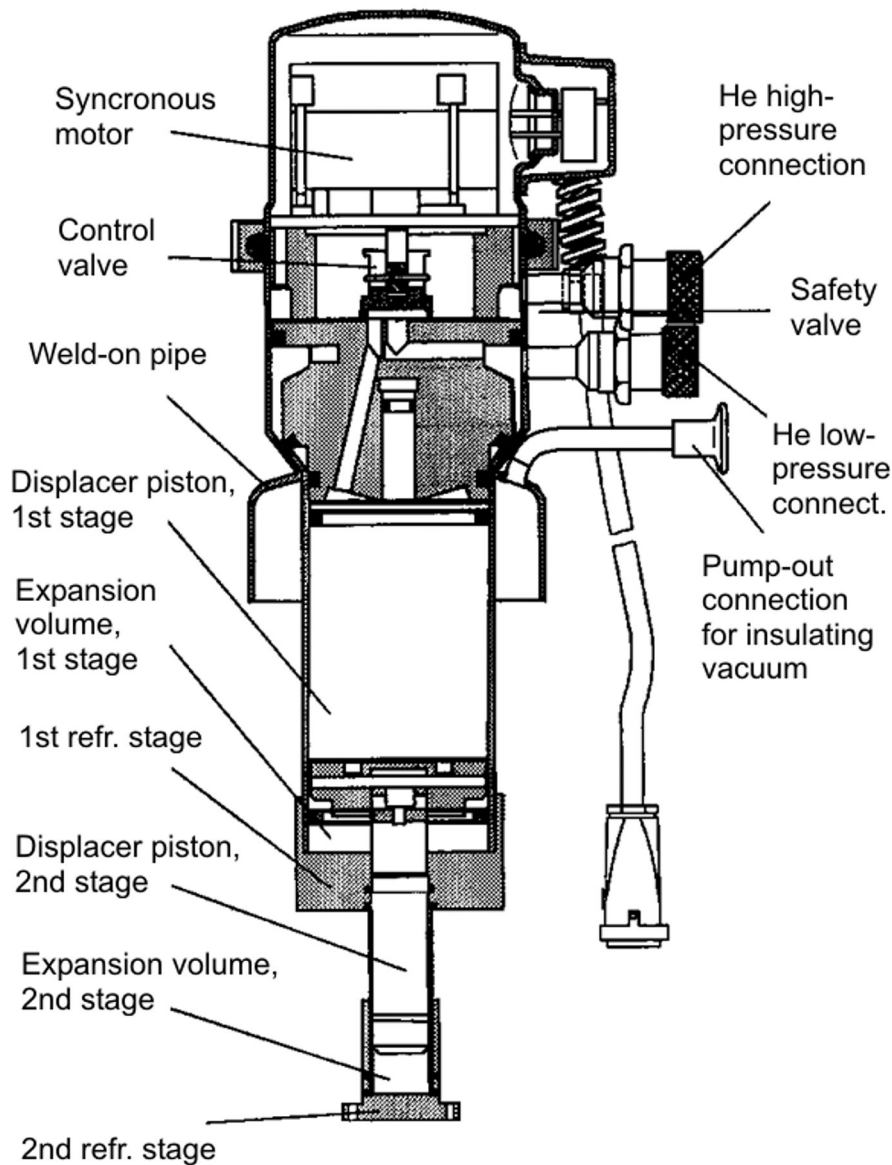
7.3 Cold head

7.3.1 General

Heat radiation plays a most important role when taking into consideration the heat impact to cryo shields and insulation foils in our magnets. This heat impact has to be reduced as much as possible in order to keep the helium boil-off rate as low as possible.

The cold head has two temperature stages. It operates with a closed helium circuit according to the Gifford-McMahon principle. For a detailed description of this principle see the functional description of the Magnet Refrigerator under "ACCU" for the MAGNETOM VISION.

Fig. 19: Cold head schematic diagram



7.3.2 Function

The cold head generates low temperatures under 120 K, i.e. "cryogenic temperatures". It has two stages connected to cool shielding shells in the magnet. The two shields are called "80 K shield" and "20 K shield". The temperatures of the shields are checked at 24 hour intervals by the MSUP.

The cold head is connected to the compressor unit via a pair of flexlines. It is switched on and off with the compressor unit. During operation, helium gas is compressed from approx. 7 bar to about 21 bar.

This high pressure helium is fed into the cold head head via the high pressure flexline. Here the pressure is decreased from 21 bar to 7 bar. During this process, energy is picked up from the two cold shields. This energy heats up the expanding helium which is fed back into the compressor via the low-pressure flexline.

Cold head, flexlines and compressor - forming a closed helium circuit - are filled and have to be recharged with high purity 5.0 - helium, i.e. the purity must be 99.9990 % or better.

The up and down movement of the displacer, the expansion of the helium gas, and the pneumatic drive of the displacement piston is controlled by a rotating valve. This valve is mounted on the shaft of a synchronous motor which is supplied with electrical power from the compressor unit. The displacer pistons of the 1st and 2nd stage are mechanically interconnected.

The frequency of the compressor piston is 2 Hz at 50 Hz line frequency and 2.4 Hz at 60 Hz line frequency. A safety valve protects the cold head against excessively high helium pressure.

The cold head has a welded-on pipe for mounting on the system and a pump-out connection for the insulating vacuum.

8.1 Overview

The Magnet Power Supply Unit (MPS 3600) is an air-cooled, moveable service tool to energize the superconducting magnet coil to the required nominal field strength.

The MPS functions are controlled by a serial connection to the host computer or to a service PC. Additional cable connections have to be established for connecting the MPS control section to the CAN bus of the MR system, for line power supply, and for connecting the magnet coil via ramp cables.

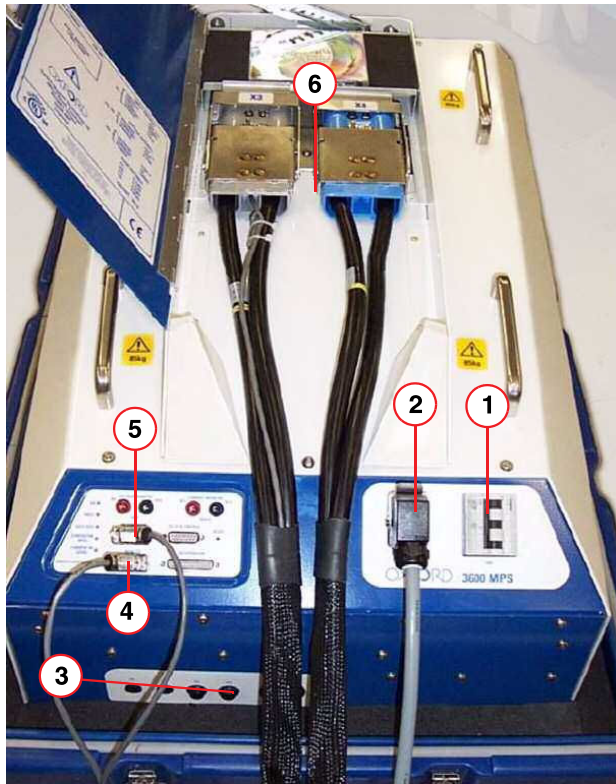
The MPS consists of:

- **MPS 3600**
- **Line power connection cable** - X1 MPS to X130 ACC roof
- **RS 232 connection cable** - X6 MPS to X12 (cable to host serial port #6)
- **CAN0 bus connection cable** - X7 to X125 ACC roof
- **Operation Manual** - Under the lid



The MPS3600 is a service tool. To ramp the OR70 magnet, the "MPSU Accessory Kit" and "MPS3600 Standard Extension Kit" are also required.

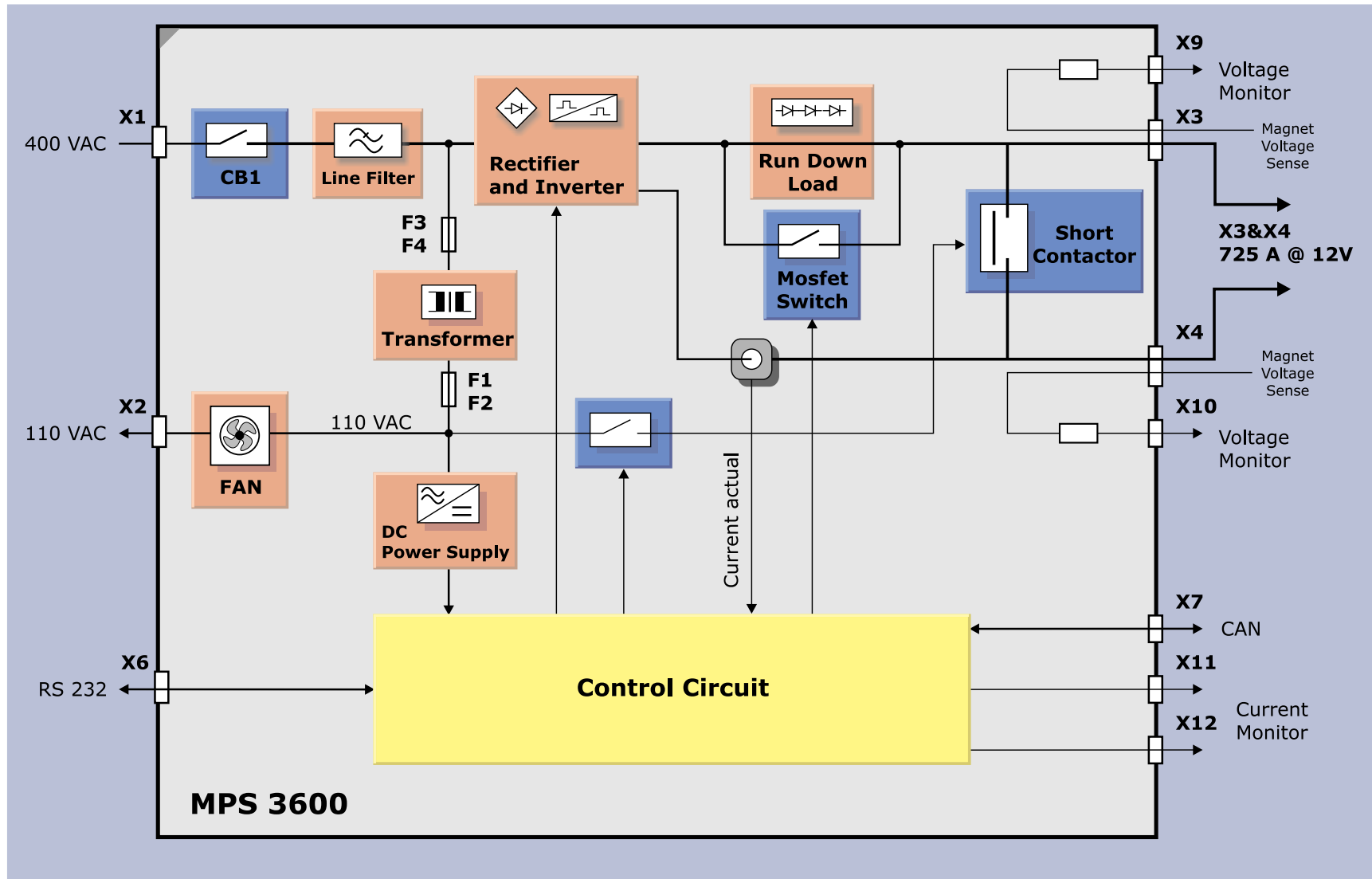
Fig. 20: MPS 3600



- (1) CB1, Mains circuit breaker
- (2) X1, MPS power connection
- (3) Fuses (F1 - F4)
- (4) X6, RS 232 connection to host or Service PC
- (5) X7 CAN0-bus connection
- (6) Ramp cables

8.2 MPS 3600

Fig. 21: MPS 3600 Function



8.2.1 Function

8.2.1.1 Power supplies

The 400 VAC 50/60 Hz 3-phase line voltage is fed via circuit breaker CB1, which routes the input supply to a line filter. Two outputs are taken from there, one via fuses F3 and F4 to a transformer that reduces the voltage to 110 VAC, the second one to the rectifier and inverter.

The power for the four cooling fans as well as for the MPS control circuits is protected by fuses F1 and F2. This 110 VAC voltage is used as energizing voltage for the shorting contactor of the main output. It is also routed to connector X2 located on the front of MPS. It can be used to connect a low-power consumption device, such as a service PC.

8.2.1.2 Ramping power supply

The 3-phase input is rectified and fed to a full bridge inverter running at about 20 kHz. The modulated square wave is fed to a 20 to 1 step down transformer that supplies a Schottky rectifier and an LC filter. The MPS output current (max. 725 A @ 12 V) is fed via a Run Down Load (RDL) to the MPS output connector X3 and X4. The RDL can be shorted via a MOSFET switch.

Actual output current is measured by a current transducer that is connected to the control circuit. For service, it can be directly measured at connectors X11 and X12.

8.2.1.3 Run Down Load, MOSFET switch

To run down the magnet, the MPS is equipped with a Run Down Load (RDL) which consists of heavy-duty diodes embedded in heat sinks and cooled by fans.

To activate the RDL, the MPS opens a MOSFET switch that is connected in parallel. The energy that is stored in the magnet coil is converted into heat inside the RDL and the magnet current is ramped down. During ramping up, the MOSFET switch is closed to directly feed the MPS current to the magnet coil.

8.2.1.4 Shorting contactor

The MPS is equipped with a shorting contactor that is normally closed. During different ramp phases, the control circuit will open the contactor if required.

The shorting contactor is opened if:

- The MPS ramps up the magnet
- The MPS ramps down the magnet

The shorting contactor is closed if:

- The MPS is switched off
- During ramping, a line power failure occurs
(Magnet current will safely run down via the ramp cables and closed shorting contactor)
- MPS performs an internal 700A self-test prior to ramping
- Magnet is captured at the beginning of the ramping-down cycle

8.2.1.5 Control circuit

The control circuit monitors and controls the operation of the MPS. Ramping commands are initiated by the operator using the host or a service PC, to be connected to the MPS. The algorithm for ramping is stored in a table at the control circuit. Required magnet-specific data are loaded from the Magnet Supervision (MSUP) via CAN bus connection (X7). During ramping, the MPS activates the switch heater by sending CAN commands to the MSUP.

If the CAN bus is disconnected, the MPS is inhibited and the fault displayed at the host or Service-PC.

8.2.1.6 Current and voltage monitoring

The MPS output current and voltage are monitored by the control circuit. It is reported via RS 232 to the host or service PC. Additionally, a DVM can be connected to the voltage monitor terminals (X9, X10) or current monitor terminals (X11, X12) on the MPS connector panel.

8.2.2 Service routines

8.2.2.1 Connecting Host or Service PC

To start ramping, the host or service PC has to be connected to X6 of the MPS (RS 232 interface). To establish communication, Hyper Terminal software is required and must be set up.

Hyper Terminal setup

Hyper Terminal has to be configured and started before the MPS is switched on.

- Select Com Port
- Baud rate = 9600
- Data bits = 8
- Parity = none
- Stop bits = 1
- Flow Control = None.
- File, properties, settings tab, Emulation = VT100



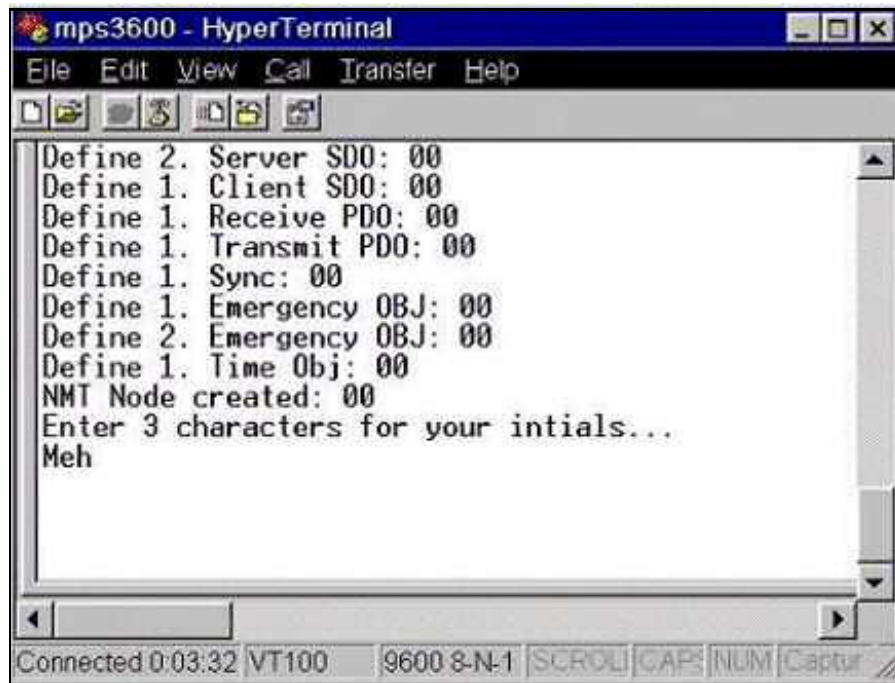
At the service PC, select the Com Port where the communication cable is connected. For host connection to Serial Distributor Port #6, always use Com 8.

Welcome screen

A welcome screen is displayed, 3 digits representing the operator (i.e. your initials) have to be entered and accepted with <ENTER>. As a next step, the magnet serial number has to be entered.

The MPS will now retrieve the magnet type, last set current, He level, date, and time from the MSUP. Once the data has been obtained, the MPSU will initiate a self test to ensure its correct functioning.

Fig. 22: Welcome screen



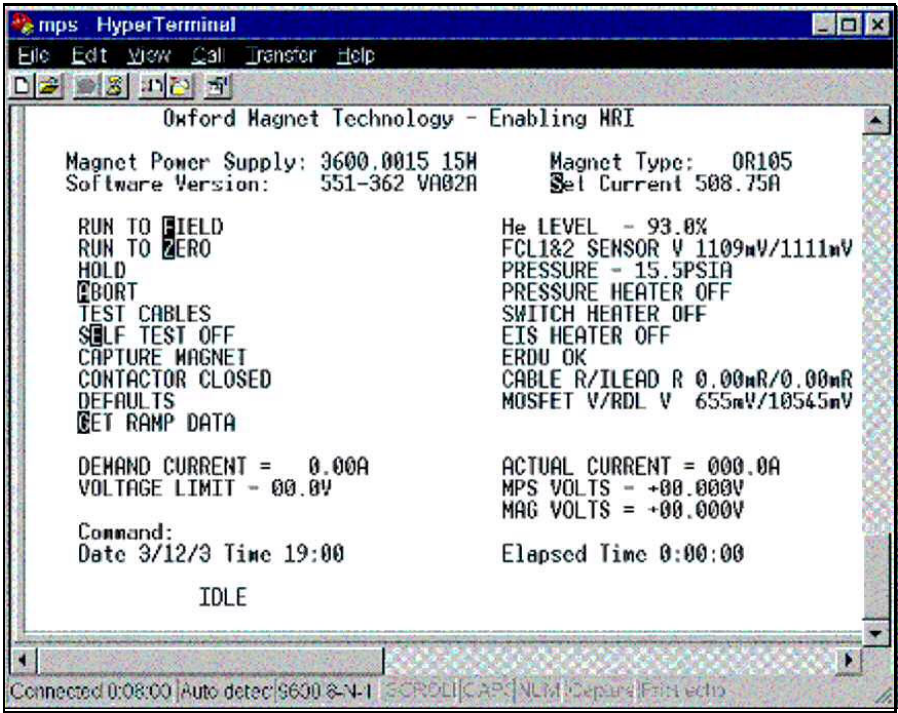
Self Test

The MPS is run up to 700 A with the shorting contactor closed. The voltage drop across the MOSFET switch is monitored. When the current reaches its peak, the voltage is recorded (V1). Next, the MOSFET switch is opened and the current runs through the Run Down Load. Before the current starts to decay, the voltage drop across the RDL is recorded at Peak Current (V2).

Operational screen

On completion of the self test, the operational screen is displayed:

Fig. 23: Operational screen (here: screen snapshot from Avanto system)



Ramping commands

Tab. 18 Ramping commands

Command	Key combination
To set magnet current:	S <space> 513.04 <Return> (Example)
To Run To Field:	F <Return>
To Run To Zero:	Z <Return>
To abort current action:	A <Return>
To refresh the screen:	O <Return>
To go to the Error Screen:	G <Return>

Errors

If a fault occurs, an error screen is displayed automatically. To access it manually, a command can be entered (see table above). The Magnet Peak Current is recorded on this screen, so if the magnet should quench, the level current at that time is stored.

MPS logfile

A logfile of the last ramp is recorded by the MPS and can be downloaded to a service PC. The following commands have to be entered:

- G <Return>, to enter the Error screen

In the Hyperterminal menu select:

- Transfer
- Capture Text
- Browse
- Select the target directory and file name: e.g. 3600 Ramp Data\OR70#9RAMP1
- Start
- L <Return> (a data screen will appear)

In the Hyperterminal menu select:

- Transfer
- Capture Text
- Stop

To return to the operational screen, press <Return>.

8.2.2.2 Run to field

Prerequisites for ramp to field:

- The MPS must be connected properly.
- Wait at least 1 hour after completing any previous ramp, or aborted ramp (including cable tests).
- Following a helium fill, wait at least 2 hours before ramping.
- Helium level must be > 50 %.

MPS system test:

- MPS performs a helium level sample
- A confirmation window pops-up if the magnet has not been filled with LHe during the last 2 hours
- MPS checks to ensure current leads are not opened or shorted

Cable test:

- Inverter and MOSFET switch is turned on, shorting contactor open
- The current in the cables is ramped up to 585 A
- MPS checks if the voltage drop across current probe (ECL) contacts is < 125 mV
- Current in the cables is ramped down to 0 A
- Calculation of cable lead resistance

Ramp to field:

- Switch heater is turned on

- MPS checks if magnet is running down already. If so, the magnet is captured and ramped down safely
- Shorting contactor is opened
- MPS demand current is set to the entered current value + 3 A
- MPS checks if entered current value + 3 A is reached (overfielding)
- Waiting time of 15 seconds
- Ramping to the entered current value
- MPS turns off switch heater and waits 2 minutes
- Ramping down cables to 0 A, MPS waits 2 minutes
- MPS turns EIS off

Ready!

8.2.2.3 Run to zero

Prerequisites for ramp to zero:

- The MPS, ECL, and cables must be connected properly
- Wait 1 hour after any previous ramp
- After helium fill at field, wait at least 2 hours before ramping
- Helium level must be > 35 %.

MPS checks prior to ramping down:

Same as checks prior to ramping up.

Cable test:

Same as for ramping up.

Ramp to zero:

- Switch heater is turned on
- Magnet capture is performed
- Shorting contactor is opened
- MPS sets demand current to 0 A and waits 10 seconds
- MPS checks if actual current is 0 A, switch heater is turned off, waiting phase of 2 min.
- EIS heater turns off

Ready!

- » Section "Magnet Monitoring", ERDU Battery Test - description of warning and alarm levels improved (CHARM MR_00376059).

There are no Hazard IDs in this document.

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